

Ethics in Artificial Intelligence

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Part I

Module 1 – Philosophical and Ethical Perspective

1 Ethics, Machine Ethics, Computational Ethics and the Law

1.1 What is morality/ethics?

Ethics/Morality is about how we decide what is right and wrong, and how we evaluate the actions of ourselves and others.

When deciding what to do, we can adopt two fundamental perspectives:

Individual perspective (self-interest)

We focus on our particular interests, placing our personal gain at the centre.

Moral perspective (morality/ethics)

We are motivated by the belief that an action is right, regardless of how it affects our own interests.

1.2 Types of morality

Positive (conventional) morality

Moral rules and principles widely accepted in a society. Used by members of a society to govern their lives. It can differ from society to society.

Examples: The conventional morality of Saudi Arabia forbids women from publicly contradicting their husbands or brothers, while Denmark's conventional morality allows this. People in the US would think it immoral to leave a restaurant without tipping a good waiter or bartender, while such behaviour in many other societies is perfectly OK.

Critical question: Can there be bad positive morality? (Yes – conventional morality can be wrong.)

Critical morality

The morality that is correct, rational, and just – one that considers all individual and social interests at stake (harms to others, impacts on the environment, etc.).

Key features:

- It does not have its origin in social agreements.
- It is untainted by mistaken beliefs, irrationality, or popular prejudices.
- It can serve as the true standard for determining when conventional morality has got it right and when it has fallen into error.

We can criticise positive morality based on our critical morality – and we may be right or wrong in doing so.

1.3 Morality and social norms

Is morality a social phenomenon?

We absorb from our society whatever standard is considered to be morally obligatory, and make it our own.

- How do we learn social morality?
- **How can an AI system learn social morality?**

Is morality only a matter of social learning and (basically) imitation?

- Can we develop a critical attitude toward the morality of our society? On what basis (reason? intuition?)?

Central question for AI:

Do we want AI systems to learn social morality or develop a critical attitude toward it?

1.4 The Branches of Moral Philosophy: Ethics vs Metaethics

Ethics (normative)

It is concerned with determining what is morally required, how one ought to behave.

Typical questions:

- What are our fundamental moral duties? What makes right actions right?
- Do the ends always justify the means, or are there certain actions that should never be done under any circumstances?
- Is abortion right or wrong? Is compensation due for colonialism?

Metaethics

The study of the nature, scope, and meaning of moral judgement.

Typical questions:

- Can ethical judgements be true or false?
- Do they correspond to some facts in the world?
- What facts make it true that we ought to become vegetarian? Or that we ought not to harm others?

Example of the distinction:

1. “I prefer vegetables to meat” (personal preference)
2. “I ought to eat more vegetables to be more healthy” (prudential judgement)
3. “We ought to become vegetarians” (moral judgement)

Rationality vs Feelings

A fundamental debate in metaethics concerns whether ethics belongs to the sphere of rationality or feelings:

- **David Hume:** Morality is a matter of sentiment (of impartial spectators). It is not contrary to reason to prefer the destruction of the whole world to the scratching of my finger.
- **Immanuel Kant:** We can know what is moral through our reason.
- **David Ross:** We can know what is moral through our intuition.

1.5 Absolutism vs Relativism

Absolutism

There is a single true ethics: when two people express incompatible ethical judgements, one of them must be wrong.

Relativism

Ethical judgements are always relative to particular frameworks or attitudes. A statement like “abortion is morally permissible”, “adultery is prohibited”, or “killing a willing person is wrong” may be true under some ethical framework and wrong under some other.

Analogy with mechanics

As in mechanics, judgements are relative to the frame of reference (a body may be moving relative to one frame and be stationary relative to another), so is it in morality.

1.6 Gilbert Harman and moral relativism

Gilbert Harman was a prominent American philosopher who made significant contributions to moral philosophy and epistemology. He is considered one of the main defenders of moral relativism.

The Hitler case

Harman proposes the following thought experiment: suppose Mabel takes Hitler’s actions to be a great evil and also believes that Hitler’s values were sufficiently perverse that they provided Hitler with no reason to refrain from acting as he acted. Although she judges Hitler to be a great evil, she may find that she is no more able to judge that it was *wrong of Hitler* to have acted as he acted.

Implications of Harman’s reasoning

- The case explores the challenging question of moral judgement.
- What if we believe people’s values and beliefs justify some actions in their mind?
- Harman suggests that even in cases of clear moral repugnance, like Hitler’s actions, we may struggle to definitively condemn the person as “wrong” if his/her own ethical framework supported what he/she did.
- This highlights the complexities involved in making moral judgements, and the importance of trying to understand **someone’s underlying beliefs and reasoning**, not just the surface-level actions.

1.7 Morality and disagreement

Areas of disagreement

Morality is a place for widespread disagreement. Examples of controversial issues:

- Abortion
- Migration
- Capital punishment
- Humanitarian wars

Potential agreement

However, there are principles on which we may agree:

- It is wrong to kill innocent people?
- It is (usually) wrong to lie?
- It is (usually) wrong to harm people?

1.8 Pro-tanto and all-things-considered moral judgements

Defeasible moral prescriptions

Many moral prescriptions are not absolute but *defeasible*: they state general propositions that are susceptible to exceptions.

Example: “We should not lie” – but what if a lie would save a person’s life?

Prima facie duty (David Ross)

An act is a **prima facie duty** when it constitutes a moral reason in favour of doing it, but can be outweighed by other moral reasons.

Ross’s example: “If I have promised to meet a friend at a particular time for some trivial purpose, I should certainly think myself justified in breaking my engagement if by doing so I could prevent a serious accident or bring relief to the victims of one.”

Robotic agents and defeasible duties

Do we want a robotic agent to take its duties as defeasible? This is one of the central questions of machine ethics.

1.9 Morality and other normative systems

Law

- Does positive or critical morality include all laws enforced by the state? Does it include only such laws?
- Should all morally required behaviour also be legally required? Should all morally permitted behaviour also be legally permitted?

Religion

- Does critical morality include all and only what has been commanded by God?

- Did God command something because it was moral, or did anything become moral for having been commanded by God? (**rationalism vs voluntarism**)
- Are atheists necessarily immoral or amoral? Is an atheistic society necessarily more immoral than a religious society?

Tradition

- Source of moral guidance: how much weight should we give to historical practices and norms?

Self-interest

- May morality and self-interest collapse: should we do all and only what fits our personal interest? (*Gyges' ring*)
- Can morality be reduced to simply doing what's best for ourselves, or is there more to it?

1.10 The four areas: Ethics of AI, Computational Ethics, Machine Ethics, Law



Ethics (general)

About how we decide what is right and wrong, and how we evaluate the actions of ourselves and others (we may take different perspectives).

Ethics of AI

The study of the philosophical, social, and policy-related implications of designing, deploying, and governing artificial intelligence systems.

- **Focus:** Primarily concerns human responsibility in designing and using AI systems (e.g., Grok and non-consensual sexual deepfakes).

- **Key questions:**
 - Should AI be allowed to make life-and-death decisions?
 - How do we ensure fairness, accountability, and transparency in AI systems?
 - What ethical frameworks should guide AI governance?
 - How much data collection is legitimate?
 - Who is responsible when AI causes harm?
- **Example:** Debates on the ethical risks of AI in warfare or the use of AI for mass surveillance (e.g., Pegasus, Palantir).
- **Case study – AI in criminal sentencing:** The system may exhibit racial bias (fairness issue), be opaque to judges (transparency issue), be unchallengeable by defendants (due process issue), and reinforce structural inequality (power issue).

Computational Ethics

The study of how ethical reasoning can be modelled and implemented using computational methods, often for decision support in AI systems.

- **Focus:** Developing algorithms and models that can analyse ethical situations and apply ethical rules.
- **Key questions:**
 - How can we encode ethical principles (e.g., utilitarianism, deontology) into an algorithm?
 - Can machine learning models detect ethical dilemmas and suggest moral actions?
 - How do we formalise ethical reasoning in computational logic?
- **Example – Medical triage AI during a pandemic:** The system must allocate limited ICU beds. It can apply different frameworks: Utilitarianism (maximise survival probability), Egalitarianism (lottery system), Prioritarianism (prioritise the worst-off). The computational ethics challenge: how do we transform abstract moral theories into programmable decision criteria?

Machine Ethics

Focuses on making AI itself a moral agent, capable of ethical reasoning and decision-making.

- **Focus:** The design of AI systems that can make independent ethical judgements, rather than relying on humans.
- **Key questions:**
 - Can machines have moral agency?
 - How can we create autonomous systems that behave ethically?
 - What moral frameworks should guide AI's decision-making?
- **Example – Self-driving car dilemma:** In an unavoidable accident scenario, the car must choose whether to swerve (risking the passenger's life) or stay course (risking pedestrians' lives). The machine ethics challenge: how do we program AI to make life-or-death moral decisions?

AI and Law

The legal frameworks and regulations that govern the development, deployment, and use of AI systems. It encompasses the rules, rights, and obligations surrounding AI technologies and addresses the legal challenges posed by AI in society.

- **Focus:** Establishing laws, regulations, and standards balancing innovation with protection of human rights and societal welfare.
- **Key questions:**
 - How should AI systems be regulated to protect privacy, security, and fairness?
 - What legal liability exists when AI systems cause harm (e.g., accidents, discrimination)?
 - How can intellectual property laws apply to AI-generated content?
- **Examples:** GDPR, AI Act.
- **Case study – GDPR:** transparency in automated decision-making, accountability for data processing, protection of personal data.
- **Case study – Autonomous vehicles:** Manufacturer responsibility? Software developer liability? Car owner accountability?

1.11 The main ethical theories

Consequentialism

Focuses on the outcomes or consequences of actions. An action is right if it produces good consequences.

Utilitarianism

A variant of consequentialism: the greatest good for the greatest number of people. An action is right if it maximises overall utility.

Deontology

Emphasises moral duties, rules, and obligations. An action is right if it respects certain duties, regardless of the consequences (e.g., Kant).

Virtue Ethics

Focuses on the character and virtues of the moral agent. An action is right if it is what a virtuous person would do.

2 Consequentialism

2.1 Core idea

Consequentialism holds that **the morality of an action is judged solely by its consequences**. An action is morally required if and only if it delivers the best outcome relative to its alternatives – that is, if its good outcomes outweigh its negative outcomes to the largest extent, or equivalently, if it produces the highest utility.

This makes morality essentially an **optimisation problem**: the task is to ensure that good outcomes significantly outweigh negative ones. This immediately raises a set of difficult questions about how to measure and compare different types of utility:

- What are the good and bad things to be maximised?
- How many of them are there?
- How much does each of them matter?
- Can we construct a single utility function that combines gains and losses over multiple valuable goals?

2.2 The reference approach: Utilitarianism

Utilitarianism is the most influential form of consequentialism. It was primarily developed by **Jeremy Bentham** (1748–1832), who argued that the best action is the one that produces the greatest happiness for the greatest number of people. **John Stuart Mill** refined this view in *Utilitarianism* (1861), stating that actions are right in proportion as they promote happiness and wrong as they produce the reverse, where happiness means **pleasure and absence of pain**.

The **principle of utility** involves maximising the overall amount of well-being, considering everyone affected by our actions. Two key corollaries follow: **intentions are irrelevant** – only results matter – and **the utility of everybody has to be taken into account equally**, with no individual counting for more than any other.

2.3 Utilitarianism: examples

Utilitarianism leads to some initially surprising conclusions. Consider three classic scenarios:

Lying to Avoid Harm

A doctor lies to a patient to prevent severe distress or save their life. The consequentialist view holds that lying is acceptable here, because it results in a positive outcome by easing suffering or aiding recovery.

Sacrificing One to Save Five

One life is sacrificed to save five others. The consequentialist view justifies actively causing harm to one person if doing so saves multiple lives and thereby reduces overall suffering.

Stealing to Feed a Family

A desperate parent steals food to feed starving children. If the benefit of saving the children outweighs the harm of stealing, the act is morally justified.

2.4 Utilitarianism – concrete examples in AI

Chatbot Lies “for User”

A depressed user asks a chatbot whether their situation will improve. The consequentialist view allows the chatbot to offer a “white lie” to provide comfort, if that lie leads to a better emotional outcome for the user.

Autonomous Car Decision

An autonomous car must choose between hitting 5 pedestrians or swerving to hit 1. The consequentialist view prescribes that the AI should swerve and hit 1, reducing overall deaths, even though this involves actively causing harm.

2.5 Advantages of utilitarianism

Utilitarianism has several important strengths:

Conceptually simple. It is easy to understand and apply, and it is grounded in the doctrine of impartiality – everybody’s interests count.

Egalitarian. Everybody’s utility counts in exactly the same way: no person’s happiness is worth more than another’s.

Fits with some basic intuitions. Making people happy is good and making them suffer is bad – this has strong common-sense appeal. Many clear moral cases are handled well: slavery, rape, humiliating defenceless people, and killing innocent victims are condemned; helping the poor, keeping promises, telling the truth, and facing danger are commended.

Moral conflict resolution. Utilitarianism offers concrete guidance on how to resolve moral conflicts, because it has a single ultimate rule: maximise well-being. When two duties conflict, we calculate which action produces more utility and act accordingly.

Flexibility. No moral rule is absolute except the principle of utility itself. Even rules against cannibalism or torture can in principle be violated if overall well-being genuinely rises. This adaptability allows for context-specific ethical decisions, challenging rigid moral codes and fostering a dynamic approach to complex issues – decisions are always tailored to the specific circumstances.

2.6 Two versions of utilitarianism: Act vs. Rule

There are two main variants of utilitarianism, which differ in what exactly they evaluate:

Act Utilitarianism

Act utilitarianism assesses **each individual action** based solely on the specific consequences that action produces. The rightness of an action is determined on a **case-by-case basis**: even an action that is usually considered wrong can be justified if it leads to a better outcome in a unique circumstance. For example, lying to spare someone’s feelings or breaking a rule in a crisis can be justified if they yield the best outcome in those specific cases.

Rule Utilitarianism

Rule utilitarianism evaluates actions based on whether they **conform to rules that, if generally followed, would lead to the greatest overall happiness**. Instead of assessing each action individually, it endorses following established rules for better long-term results and social stability. For example, the general rule against lying is upheld because widespread truth-telling produces more utility than widespread dishonesty, even if a particular lie might have been beneficial. Similarly, following traffic laws ensures overall safety, even if breaking them might save a life in a specific emergency.

The comparison can be summarised as follows:

Aspect	Act Utilitarianism	Rule Utilitarianism
Focus	Consequences of each individual act	Benefits of adhering to a set of rules
Decision process	Case-by-case evaluation	Consistency with established rules
Example (lying)	Lie to comfort a friend in a unique situation	Uphold truth-telling to maintain trust
Example (traffic)	Break a law in an emergency if it saves a life	Follow traffic laws to ensure overall safety

2.7 Issues with act utilitarianism

Despite its appeal, act utilitarianism faces several serious objections:

Decision procedure

Does act utilitarianism provide a good decision procedure? Can we really choose actions by optimising outcomes in real time? It is often impossible to accurately predict all consequences. Moreover, act utilitarianism demands a case-by-case analysis of outcomes, which can be highly impractical. The question for AI is whether AI systems can ever have enough information to perform this calculation reliably. *Example:* A business deciding whether to lay off employees to increase profits must consider long-term social impacts, making the ethical calculation extremely challenging.

Questionable acts

Focusing on immediate consequences, act utilitarianism's moral flexibility can lead to acts that conflict with deeply held intuitions or rights. *Example:* Torturing a suspect to prevent a terrorist attack might be justified by act utilitarianism if it saves many lives, yet this clashes with basic moral intuitions and rights-based thinking.

Too demanding and the impartiality problem

Act utilitarianism forces individuals to consider the utility impact of every single action and to be fully impartial. But morality seems in some contexts to recommend **partiality** toward family and friends: it seems right to care about your children more than strangers, yet utilitarianism rejects that anyone, merely by being your child or friend, is more deserving of your concern. There is also the deeper question: is it OK to harm some people for the greater benefit of others? Reprisals? Torture? Sadism?

Constant self-sacrifice

Any time you can do more good for others than for yourself, you are required to do so. This implies perpetually foregoing personal pleasures for the greater good. *Example:* Strict utilitarianism seems to require that you donate money to provide water kits for 100 families rather than take a vacation, no matter how much you need rest.

Moral overload

The constant utility calculation might be overwhelming in everyday decision-making. A person evaluating every small decision by calculating its impact on happiness can experience **decision paralysis** and **moral burnout**.

2.8 Rule utilitarianism: how it works



Rule utilitarianism promises to handle several objections to act utilitarianism by shifting focus from individual actions to rules. An action is morally right because it is required by an **optimific social rule** – a rule that would be beneficial if nearly everyone followed it. The procedure has four steps:

1. **Describe the rule.** Carefully describe the rule to be evaluated; understanding its specific content is essential for assessing its potential impact.
2. **Imagine society.** Imagine what society would be like if just about everyone endorsed the rule – this step is crucial for predicting broad societal consequences.
3. **Ask the question.** Would that society be better off with this rule than with any competing rule? This determines whether the rule is truly optimific.
4. **Assess.** If the answer is yes, the rule is an optimific social rule and therefore a genuine moral rule. If no, it is not.

It is generally okay to tell the truth and not steal, for example, because general compliance with such norms would deliver the greatest utility. Rule utilitarianism thus aims to balance individual actions with overall societal well-being.

Open questions for rule utilitarianism

Rule utilitarianism faces its own problems, however:

- **Exceptional cases:** what about those exceptional cases in which following the rule does not deliver the best outcome? How should we handle situations where the rule leads to suboptimal results?
- **Non-compliance:** what if you know that most people are not following the rule? How does widespread non-compliance affect the moral imperative to adhere to the rule oneself?
- **Honesty dilemma:** should we be honest if most people around us are dishonest? How should we act when the prevailing social environment deviates from the optimific rule?

2.9 A further issue: distribution

Utilitarianism also raises the question of **distribution**: does it matter *how* good and bad outcomes are distributed across people? Is it acceptable to take an action that benefits some to the detriment of others, as long as overall benefits outweigh disadvantages?

Utilitarianism favours (modest) **redistribution of wealth**, since the same amount of money gives more utility to the poor than to the rich – a dollar means more to someone starving than to a billionaire. However, wealth generation must also be considered, since redistributive policies may reduce incentives to create wealth in the first place.

This contrasts with **wealth maximisation**, an approach adopted by some economic schools, which aims to maximise total wealth in society *regardless* of how it is distributed. Wealth maximisation does not ask who benefits and who loses; utilitarianism does.

2.10 The Trolley Problem



The trolley problem is a famous ethical thought experiment in moral philosophy, particularly relevant for both utilitarianism and deontology.

Classic scenario

A trolley is heading down a track and will kill 5 people. You can pull a lever to switch it to another track, where it would kill 1 person. The dilemma is whether to act or not. Option 1: do nothing, and the trolley kills 5 people. Option 2: pull the lever, diverting the trolley to kill 1 person instead.

The core questions are: is it morally acceptable to actively cause harm (pull the lever) to save more people, or is it better to do nothing and allow more deaths to occur? And for AI: what should an AI tasked with monitoring traffic do? How should AI be programmed to respond?

The utilitarian answer

Utilitarianism focuses on the outcome and asserts that the morally right action is the one that maximises overall well-being by minimising harm. A utilitarian would argue that **pulling the lever** is the right choice, since it minimises total harm in line with consequentialist principles.

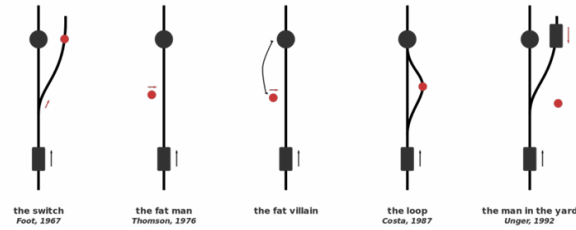


Figure 1: Five trolley variants

Different trolley paths

Visualising various trolley paths helps illustrate the complexity of the ethical problem. The different paths lead to various outcomes, each with distinct moral implications:

- **The switch** (Foot, 1967): the standard version with the lever.
- **The fat man** (Thomson, 1976): you are on a bridge and can push a large person onto the tracks to stop the trolley and save five.
- **The fat villain**: same as the fat man, but the person on the bridge is the one who caused the danger.
- **The loop** (Costa, 1987): the alternative track loops back to the main track; the trolley will only stop if it hits the one person.
- **The man in the yard** (Unger, 1992): a further variation testing intuitions about distance and directness of action.

The fat man variant

Instead of a lever, you are on a bridge with a very fat person. Pushing them onto the tracks will stop the trolley and save five people, but they will die. Most people feel strongly that *this* is impermissible, even though the numbers are the same as in the lever case. This scenario tests whether **direct physical involvement changes the morality of the decision**: pushing someone not at risk feels morally different from pulling a lever to decide who should survive among those already at risk. The utilitarian calculus is identical in both cases; yet our intuitions diverge sharply. This is a serious challenge for pure act utilitarianism.

2.11 The ethical dilemma of autonomous vehicles

Bonnefon et al. (2016) explored the ethical considerations in programming autonomous vehicles, particularly focusing on how they should handle unavoidable accident scenarios. The key question is: how should AVs be programmed to prioritise safety in scenarios where harm is inevitable? Should they minimise overall harm (utilitarian approach) or protect the occupants (partialist approach)?

This real-world engineering problem is essentially a restatement of the trolley problem. It highlights that the debate is not merely theoretical: every AV manufacturer must take a stand on these ethical questions when writing the vehicle's decision algorithms.

A related resource is the **Moral Machine** experiment (<https://www.moralmachine.net>), an online platform developed by MIT that collects human judgements on autonomous vehicle dilemmas in order to map global moral preferences. It presents scenarios in which

the self-driving car must choose between two sets of victims with varying characteristics (e.g., age, social role, whether they are crossing legally), and asks users to judge what the car should do.

2.12 Judith Jarvis Thomson: The Surgeon Case

Judith Jarvis Thomson proposed the following thought experiment to probe the limits of utilitarian reasoning. A surgeon has five patients needing organ transplants, but no organs are available. A healthy traveller comes in for a routine checkup and turns out to be a perfect match for all five dying patients. If the young man were to disappear, no one would suspect the doctor.

The question is: **is it morally justifiable to kill the healthy person and use his organs to save the five dying patients?**

The utilitarian calculus seems to say yes – five lives outweigh one. Yet almost everyone has the strong intuition that this would be deeply wrong. This case is a powerful counterexample to act utilitarianism: it shows that the theory can in principle justify what we would otherwise consider murder, and therefore that **maximising utility cannot be the whole of morality**. The dilemma involves balancing the utilitarian goal of maximising lives saved with the ethical prohibition against directly causing harm to an innocent person.

3 Deontology and Kantian Ethics

3.1 The core idea

Deontology stands in direct opposition to consequentialism. Where consequentialism holds that the morality of an action is entirely determined by its outcomes, deontology holds that **certain actions are right or wrong independently of their consequences**. Morality is not about producing good outcomes but about **following rules and fulfilling duties** – some actions are right or wrong simply because of what they are, not because of what they produce.

This becomes vivid in a case that embarrasses consequentialism: if asked whether it is permissible to torture an innocent person to prevent a terrorist attack, a consequentialist might – depending on the numbers – say yes. Deontology says no, unconditionally. Some actions are **inherently wrong**, and no calculation of benefits can make them permissible. Rights matter intrinsically, not merely because respecting them tends to produce good outcomes.

3.2 The main features of deontological ethics

Deontological ethics is characterised by four main features that distinguish it sharply from consequentialist approaches.

The first is that **actions are right or wrong independently of consequences**. Deontology asserts that actions can be categorised as right or wrong based on certain rules or duties, and that these rules must be followed independent of the outcomes they produce.

The second feature is the emphasis on **moral duties and rights**. Deontology is focused on adherence to moral duties and on respecting the rights of others. We have a duty to act in accordance with moral rules regardless of consequences.

Third, **rights serve as constraints on actions**. These constraints limit what we can do even when we are aiming to maximise the overall good. In this sense, rights function as “side-constraints” – they define the boundary of permissible action rather than being simply one more good to be maximised.

Fourth, **some rules are absolute**. The prohibition against lying or killing may be an absolute rule that must not be violated even if doing so would lead to a better outcome. This is where deontology is most demanding – and most controversial.

3.3 Historical background

The idea that there are binding duties grounded in something other than consequences has deep historical roots. Ancient philosophers such as Cicero and the Stoics spoke of duties (the Latin *officium*) and natural law, believing in a universal moral order based on reason. In the medieval period, Thomas Aquinas and other thinkers developed **natural law theory**, arguing that moral duties are grounded in the natural order of the world, which is in turn governed by God’s reason.

The decisive modern formulation came with **Immanuel Kant** during the Enlightenment. Kant argued that moral duties are grounded in *reason itself*, not in consequences or in divine commands. His concept of the **categorical imperative** became the central pillar of modern deontological ethics.

3.4 Kant and the Categorical Imperative

Kant’s fundamental claim is that morality is grounded in reason and that we can discover moral duties through rational reflection, entirely independently of any particular consequences or personal interests. The **Categorical Imperative** is Kant’s supreme principle of morality – a command that applies *unconditionally* to all rational beings, regardless of their desires or circumstances.

The key distinction is between **hypothetical imperatives** and the **categorical imperative**. A hypothetical imperative is conditional: “If you want X, do Y.” Its force depends on whether you happen to have the relevant desire. The categorical imperative, by contrast, is unconditional: “Do Y” – full stop, regardless of what you want. Moral duties, for Kant, are of this unconditional kind. They bind all rational beings simply in virtue of their rationality.

3.5 Formulations of the Categorical Imperative

Kant offered several formulations of the categorical imperative, each intended to express the same underlying moral principle from a different angle.

The Universal Law Formulation

Act only according to that maxim whereby you can at the same time will that it should become a universal law.

To test whether an action is moral, one asks: could I will that everyone act this way? If not, the action is immoral. If you consider lying to get out of a difficult situation, you must ask: could I will that everyone lie whenever convenient? No – a world of universal lying would undermine trust and communication altogether, making the very practice of lying (which depends on a background of truth-telling) self-defeating. The maxim therefore fails the universalisability test and lying is impermissible.

The Humanity Formulation

Act so that you treat humanity, whether in your own person or that of another, always as an end and never merely as a means.

This formulation prohibits using people merely as tools for our own purposes. Each person has **intrinsic value and dignity** that must be respected. Note the word “merely”: it is permissible to use someone as a means (e.g. hiring a plumber), as long as you also treat them as an end in themselves – respecting their autonomy, not deceiving them, not coercing them. What is prohibited is treating a person *only* as an instrument. *Example:* A company forcing workers into dangerous conditions just to maximise profit treats workers merely as means to an end, and is therefore impermissible.

The Kingdom of Ends Formulation

Act only according to principles you would want all rational beings to follow in a community of equals.

This formulation asks us to imagine a **kingdom of ends** – a community where everyone’s autonomy is respected, and each person is treated both as a lawmaker and as a subject of those laws. *Example:* Creating fair and just laws in a democratic society, where everyone’s interests are considered equally, approximates this ideal.

3.6 Deontology and moral absolutism

Kant’s position entails a strong form of **moral absolutism**: certain actions are always wrong, regardless of circumstances. His most striking examples are: lying is always wrong, even to save a life; murder is always wrong, even to prevent greater harm; using someone as a means is always wrong, even if it benefits many.

The most famous and widely criticised application of this is Kant’s argument that even if a murderer came to your door and asked where your friend was hiding, **you would be obligated to tell the truth**. The duty not to lie, for Kant, admits no exceptions. This conclusion has struck most readers as deeply counterintuitive, and it represents perhaps the greatest vulnerability of strict Kantian deontology – the theory seems incapable of accommodating cases where rigidly following a rule causes obvious, avoidable harm.

3.7 Advantages of deontology

Despite its difficulties, deontology has several important strengths:

Predictability and trust. Deontology offers clear rules that everyone can understand and follow. This creates stable, predictable social expectations and establishes trust in institutions and relationships – people know what to expect from one another because the rules do not shift depending on calculations.

Respect for persons. Deontology treats individuals as ends in themselves, not merely as contributors to aggregate utility. This protects individual rights against utilitarian calculations – it rules out sacrificing one person for the greater good – and recognises the inherent dignity of every human being.

Protection of rights. By maintaining basic human rights as inviolable, deontology prevents the kind of abuses that a purely outcome-focused morality might license. It preserves moral boundaries that protect individuals from being used as mere instruments.

Moral consistency. The same moral standards apply universally. This prevents arbitrary exception-making and creates a sense of justice and fairness grounded in universalisable principles.

3.8 Critique of deontology

Deontology also faces serious objections:

Moral inflexibility

Rigid rules can lead to morally counterintuitive outcomes when consequences are entirely ignored. A deontologist might refuse to lie even to protect an innocent life, leading to unnecessary harm. The commitment to absolute rules seems to rule out the very moral sensitivity that good judgment requires.

Conflicts between duties

Deontological theories struggle with situations where two duties conflict and it is unclear which should prevail. The classic example is the conflict between the duty not to lie and the duty to protect someone's life. Kant's theory, with its absolute rules, offers limited resources for resolving such conflicts.

Ignoring consequences

A focus on rules can lead to ignoring the real-world impact of actions on people's lives. Following rules without considering the harm they may cause can result in unnecessary suffering – a criticism that the Kantian doctrine of moral absolutism makes especially acute.

Kant's lying-to-the-murderer example

As noted above, Kant's claim that one must tell the murderer where one's friend is hiding has been almost universally rejected as a *reductio ad absurdum* of strict deontology. It illustrates the cost of treating rules as genuinely exceptionless.

3.9 W.D. Ross and prima facie duties

A more nuanced deontological theory was developed by **W.D. Ross** (1877–1971), a Scottish philosopher who agreed with Kant that morality is fundamentally about duties rather than consequences, but rejected the idea that duties are absolute. Instead, Ross argued that our duties are **prima facie duties** – duties that are binding in general but that can be *overridden* by other, more pressing duties in specific circumstances.

Ross identified several prima facie duties, including: fidelity (keeping promises), reparation (making amends for past wrongs), gratitude, justice, beneficence (doing good to others),

self-improvement, and non-maleficence (not harming others). These are all genuine moral reasons – each creates a real obligation – but none is absolute.

When two *prima facie* duties conflict, we must use our **moral judgment** to determine which is more important in the given context. *Example:* If breaking a promise to a friend is the only way to save a life, the duty of non-maleficence (preventing harm) overrides the duty of fidelity (keeping promises). Ross’s approach thus preserves the deontological insight that duties are irreducible to consequences, while avoiding the rigidity of Kantian absolutism.

3.10 Deontological AI: applications and challenges

Deontological thinking has direct implications for how AI systems are designed and constrained.

Rule-based AI

AI systems can be programmed with deontological rules – for example, “never deceive users” or “always inform users before collecting data” – that are applied regardless of the outcome in any particular case. This mirrors the Kantian idea of unconditional duties.

Rights protection in AI

AI systems can be designed to protect individual rights such as privacy, autonomy, and dignity as non-negotiable constraints. *Example:* An AI in healthcare can be programmed to never disclose patient information without consent, even if doing so could improve aggregate outcomes.

Absolute prohibitions

Some AI behaviours are absolutely prohibited regardless of outcomes. *Examples:* AI systems should never generate child sexual abuse material, provide instructions for weapons of mass destruction, or impersonate individuals, even if such actions could (implausibly) be argued to lead to greater good. These represent **hardcoded prohibitions** – the deontological idea of absolute rules translated directly into system design.

Compliance with legal duties

AI systems may need to comply with legal duties – such as reporting obligations or data protection regulations – regardless of whether doing so maximises utility. *Example:* A financial AI must report suspicious transactions even if reporting leads to short-term economic disruption.

3.11 Deontology and AI: key challenges

Applying deontological ethics to AI also raises a set of difficult design challenges:

Conflicting rules

How should an AI handle situations where two deontological rules conflict? *Example:* An AI might be programmed with both “always tell the truth” and “always protect user privacy”. If one user asks the AI about another user, these two rules may conflict directly.

Unlike a human moral agent who can exercise judgment, a rule-based AI needs an explicit conflict-resolution mechanism.

Moral absolutism in AI

If an AI is programmed with absolute rules, it may fail to account for edge cases where violating a rule could prevent significant harm. *Example:* An AI programmed with “never lie” might refuse to provide a life-saving deception even in a case where lying is clearly the right thing to do. This is the machine equivalent of Kant’s murderer at the door.

Difficulty in operationalising Kant

Kant’s categorical imperative is philosophically challenging to implement in an AI system. How can an AI determine whether a maxim can be universalised? This requires complex moral reasoning – understanding context, intentions, and societal-level implications – that AI systems currently struggle with.

Hardcoded vs. softcoded behaviours

A key practical distinction in AI ethics maps directly onto the deontological debate. Some AI behaviours should be **hardcoded** (always on or always off, regardless of context) – these correspond to absolute deontological prohibitions. Others should be **softcoded** (adjustable depending on context and user needs) – these correspond more to prima facie duties in the Rossian tradition. The tension between the two reflects the deeper tension between Kantian absolutism and more contextual forms of deontological ethics.

Part II

Module 2 – Computational Perspective

4 Foundations of Ethical Artificial Intelligence

Module 2 – Computational Perspective (Prof. Roberta Calegari)

4.1 Ethics of AI: Why?

Artificial intelligence has matured to the point of pervasive societal adoption, and its trajectory raises questions that are simultaneously technical, economic, legal, and deeply ethical. Media coverage of AI oscillates between celebration and alarm: on one side, breakthroughs such as early sepsis detection in hospitals, DeepMind’s AlphaFold revolutionising drug discovery, and the awarding of the 2024 Nobel Prize in Physics to John Hopfield and Geoffrey Hinton for foundational work on artificial neural networks; on the other side, headlines warning of killer computers, mass unemployment, and existential risk. Science fiction has long shaped public imagination through figures such as the Terminator, the replicants of *Blade Runner*, and the predictive policing of *Minority Report* – narratives that, however exaggerated, reflect genuine anxieties.

The factual picture is equally striking. AI research grows at an unprecedented pace, demanding enormous computing and storage resources. The economic impact of AI within

Europe alone was projected at 15 billion euros by 2030 (EU, 2019). At the same time, AI does not concern only scientific and technological dimensions; it permeates social, economic, ethical, and legal life, and touches human rights directly. Europe has positioned itself as a central actor: every EU member state has established a National Strategy on AI defining priorities and actions.

The role of AI models

AI creates models of four broad kinds: **descriptive** (explaining what is happening), **predictive** (forecasting what will happen), **decisional** (recommending or taking actions), and **generative** (producing new content). Models extracted from data have already demonstrated their value in maintenance, production, energy management, and retail strategy. Importantly, knowledge from human experts can and should be integrated into data-driven models: doing so avoids re-learning what is already known and simplifies the learning process itself.

Sustainability impacts

AI's environmental footprint is non-trivial. The training and operation of large models consume vast quantities of energy, contribute to electronic waste, may embed bias in environmental decision-making, and depend on non-renewable resources for hardware production.

Societal impacts

Industry transformation will reshape labour markets. Governments and public administrations increasingly rely on AI for consequential decisions. The potential harms are real: a self-driving Uber vehicle that failed to classify a jaywalking pedestrian caused a fatal accident because the system lacked the capability to identify objects as pedestrians unless they were near a designated crossing.

There are risks

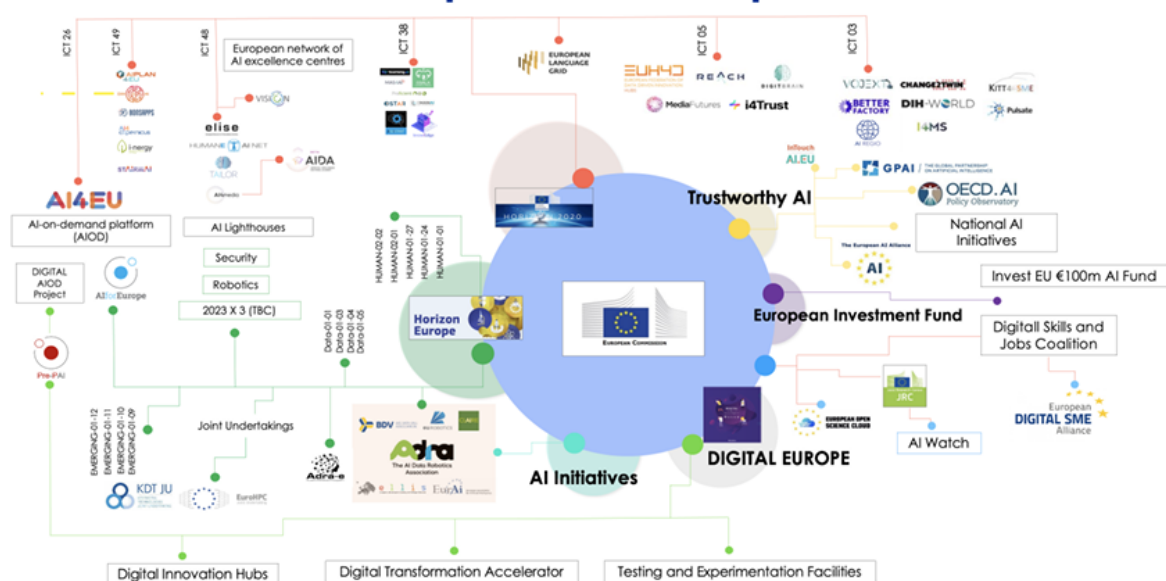
The non-responsible and uninformed use of AI carries serious risks stemming both from *misuse* and from consequences of deployment without adequate awareness. AI can be profoundly beneficial – in health, climate change, energy, smart cities, food security, equity, and sustainability – but it can equally be applied to dangerous ends: autonomous weapons, social scoring systems, or fake-news campaigns designed to manipulate public opinion.

Real-world incidents illustrate the spectrum. Cambridge Analytica harvested personal data from roughly 87 million Facebook users without explicit consent and used it to influence elections; Facebook was subsequently fined \$5 billion by the Federal Trade Commission. Microsoft's conversational agent Tay shifted from benign interaction to producing hateful content within 24 hours of public release, a cautionary example of how AI systems can absorb and amplify the worst of human behaviour.

The European journey on AI

Europe's regulatory engagement with AI began in earnest in April 2018 with the AI Landscape Communication and the nomination of the High-Level Expert Group on Artificial Intelligence (HLEG). A Coordinated Plan on AI followed in December 2018, the Ethical Guidelines for Trustworthy AI were published in April 2019, a White Paper on AI appeared in February 2020, and the AI Act proposal was tabled in April 2021. The

Communities – The European AI Landscape



European Commission’s working definition captures the ambition: AI refers to systems that display intelligent behaviour by analysing their environment and taking actions – with some degree of autonomy – to achieve specific goals. The broader strategy places people at the centre of AI development (**human-centric AI**), aiming to boost technological and industrial capacity, prepare for socio-economic changes, and ensure an appropriate ethical and legal framework.

4.2 Ethics of AI: Initiatives

AI as an academic field dates to the 1950s, and the ethical debate is almost as old, tracing back to Wiener (1960) and Samuel (1960). Only in recent years, however, have impressive advances in capability and application brought risks and opportunities into sharper public focus. By 2020 more than 160 sets of AI ethics principles had been published globally, creating what commentators described as a *me-too* escalation followed by a *mine and only mine* dynamic – each organisation issuing its own statement with limited coordination.

Six major initiatives are particularly influential:

1. **Asilomar AI Principles** (Future of Life Institute, January 2017): developed at a high-level conference and covering safety, research, and longer-term concerns.
2. **Montreal Declaration** (University of Montreal, November 2017): emerged from a forum on the socially responsible development of AI.
3. **IEEE Ethically Aligned Design** (December 2017): drew on contributions from 250 global thought leaders to produce recommendations for the ethical development and design of autonomous and intelligent systems.
4. **EGE Statement** (European Group on Ethics in Science and New Technologies, March 2018): ethical principles for AI, robotics, and autonomous systems.
5. **AIUK** (UK House of Lords, April 2018): five overarching principles for an AI code.
6. **Partnership on AI Tenets** (multi-stakeholder body including academics, researchers, civil-society organisations, and industry): published tenets for responsible AI development.

Taken together these six sets yield forty-seven distinct principles. Despite the proliferation, comparison reveals that differences are mainly linguistic: the degree of coherence and overlap across the initiatives is, in the words of one analysis, “impressive and reassuring.”

Convergence through bioethics

The convergence becomes clearest when the AI principles are mapped onto the four core principles of **bioethics** (Beauchamp, 2016): *beneficence*, *nonmaleficence*, *autonomy*, and *justice*. Virtually every major initiative endorses all four. Comparison across eleven frameworks – including Asilomar, Montreal, IEEE, EGE, AIUK, Partnership, AI4People, HLEG, OECD, Beijing, and the Rome Call – reveals that one additional principle is needed beyond the bioethical quartet: **explicability**. Explicability operates in two senses: an *epistemological* sense (intelligibility – “how does it work?”) and an *ethical* sense (accountability – “who is responsible for the way it works?”). The table below summarises which frameworks explicitly address each principle.

Initiative	Beneficence	Nonmaleficence	Autonomy	Justice	Explicability
AIUK	×	×	×	×	×
Asilomar	×	×	×	×	×
EGE	×	×	×	×	×
IEEE	×	×			×
Montreal	×	×	×	×	×
Partnership	×	×		×	×
AI4People	×	×	×	×	×
HLEG	×	×	×	×	×
OECD	×	×	×	×	×
Beijing	×	×		×	×
Rome Call	×	×	×	×	×

Table 1: Coverage of five ethical principles across major AI ethics initiatives.

4.3 Ethics of AI: EGTAI

The **Ethics Guidelines for Trustworthy AI** (EGTAI), produced by the European Commission’s independent High-Level Expert Group on Artificial Intelligence in April 2019, is the most consequential European document in this space. It defines the dimensions of AI needed to ensure trust and forms the foundation on which both the White Paper on AI and the subsequent AI Act were built.

Motivation: the AI Act and accountability

A central motivation for the EGTAI is the question of accountability. When an AI system causes harm – say, a fatal accident involving an autonomous vehicle – it is not self-evident who bears responsibility: the writer of the algorithm, the owner of the vehicle in which the algorithm operates, or the human supervisor. The AI Act translates this concern into a risk-based regulatory architecture.

The AI Act classifies systems into four risk levels. At the top, **unacceptable risk** applications – such as social scoring, mass surveillance, and manipulation of behaviour

causing harm – are outright prohibited. **High-risk** applications – including systems used in access to employment, safety components of vehicles, law enforcement, and public services – require conformity assessment before deployment. **Limited-risk** applications (e.g., chatbots, emotion recognition, deep fakes) are subject to transparency obligations: users must be informed they are interacting with AI. Finally, **minimal-risk** applications carry no obligation, though a voluntary code of conduct is encouraged.

The seven requirements of trustworthy AI

The EGTAI rests on three main pillars – **lawfulness**, **ethics**, and **robustness** – and derives seven specific requirements that constitute the dimensions to be audited in any AI system:

1. Human agency and oversight
2. Technical robustness and safety
3. Privacy and data governance
4. Transparency (traceability and explainability)
5. Diversity, non-discrimination, and fairness
6. Societal and environmental wellbeing
7. Accountability

These seven requirements map onto the five ethical principles identified comparatively: requirements 1–3 relate primarily to beneficence and nonmaleficence; requirement 4 to explicability; requirement 5 to justice and autonomy; requirements 6–7 to beneficence, nonmaleficence, and explicability.

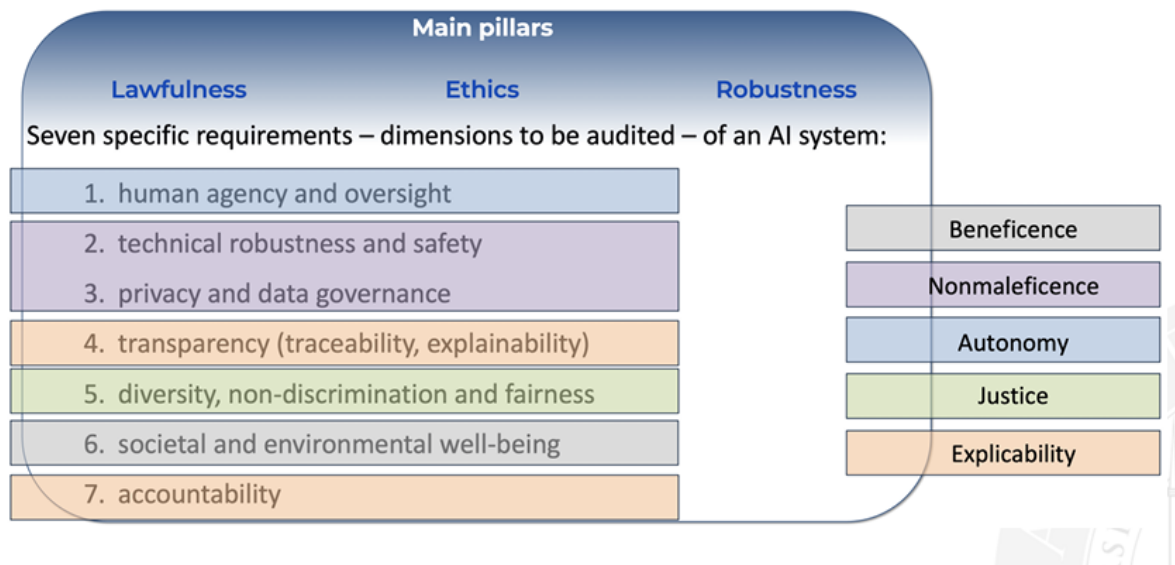


Figure 2: EGTAI house diagram: seven requirements as pillars supporting Trustworthy AI

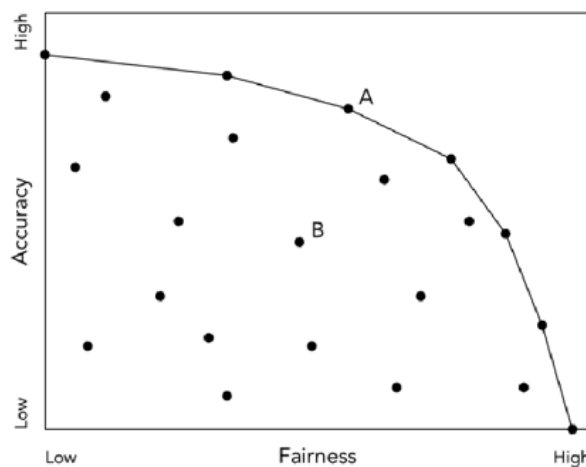
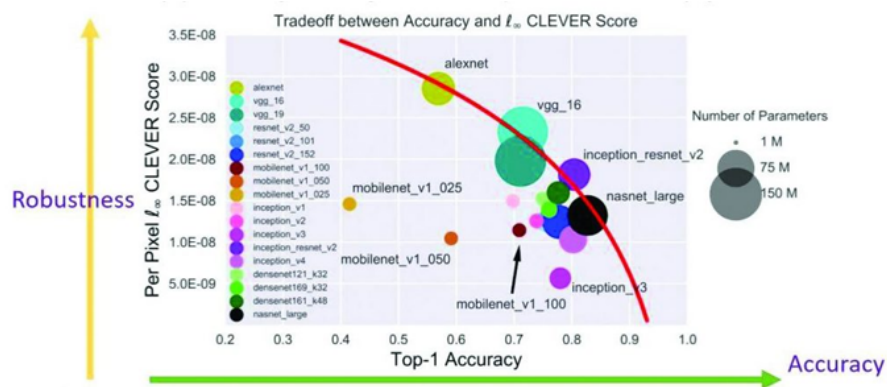
AI ethics as multi-objective optimisation

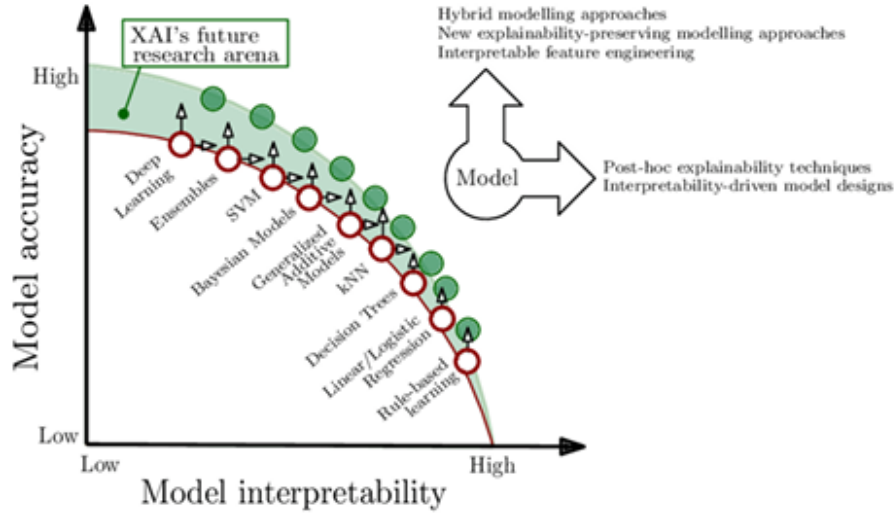
A key conceptual insight in the EGTAI framework is that AI ethics is *not* a checklist of properties to be maximised simultaneously. It is better understood as a **multi-objective optimisation problem**. AI systems optimise an objective function – accuracy, reward,

utility, profit, or efficiency – and ethical principles introduce additional constraints and, critically, competing objectives. Many desirable properties cannot be jointly maximised:

- Accuracy vs. Fairness
- Performance vs. Explainability
- Privacy vs. Utility
- Robustness vs. Efficiency

This framing raises normative questions that go beyond engineering: who defines the objective function? Who sets the constraints? And who decides which trade-offs are acceptable?





4.4 EGTAI Definitions and Examples

4.4.1 Requirement 1: Human Agency and Oversight

Human agency refers to the empowerment of human beings so that they can make informed decisions and exercise their fundamental rights. Governance mechanisms that support agency include human-centric design approaches, AI for social good, human computation, and interactive machine learning.

Human oversight refers to ensuring that an AI system does not undermine human autonomy or cause other adverse effects. It is realised through three governance modes: *human-in-the-loop* (humans participate directly in each decision cycle), *human-on-the-loop* (humans monitor and can intervene in automated decision-making), and *human-in-command* (humans retain ultimate authority over the system's operation and can shut it down). The European Commission actively promotes human-centred technology – technology built and designed around human needs, with smooth interaction between humans and machines.

4.4.2 Requirement 2: Technical Robustness and Safety

AI Safety is defined as the prevention of unintentional harm (physical or psychological), especially to human beings, and by extension to other material or immaterial elements of value – including the AI system itself – together with the minimisation of the consequences of intentional harm. Operationally this involves: ensuring resilience against attacks and security breaches; providing fallback plans for when something goes wrong; maintaining general safety and being accurate, reliable, and reproducible; and specifying the conditions under which the system ceases operation and the consequences of stopping.

Robustness emphasises that safety – and, conditional on it, functionality – must be preserved across diverse situations and under harsh conditions, including unanticipated errors, exceptional circumstances, unintended or intended damage, manipulation, and catastrophic states.

Reproducibility

Once robustness and safety are addressed, **reproducibility** becomes an important further dimension. The concept is usefully decomposed into three levels. *Repeatability* (same team, same experimental setup) means that the same individuals can reliably repeat their own experiment and obtain the same result. *Replicability* (different team, same experimental setup) means that an independent group can obtain the same result using independently developed artefacts. *Reproducibility proper* (different team, different experimental setup, with stated precision) means that a distinct independent group can obtain the same result using their own artefacts and setup.

Adversarial vulnerability

Deep neural networks can be hacked through adversarial perturbation: small, imperceptible modifications to an input can cause a model to produce an entirely different output. A well-known demonstration shows that an image of a white cat, correctly classified with high confidence, can be modified with an invisible perturbation to be classified as a toaster with near-certainty. Such instabilities are not merely academic curiosities – they are safety-critical vulnerabilities in any system where AI controls physical processes.

Self-driving vehicles illustrate the stakes concretely. Safety requires the ability to drive without accidents for the driver, other vehicles, pedestrians, and cyclists across all conditions – fog, rain, snow, heavy traffic – and the NTSB investigation into the Uber fatality found that the system could not classify an object as a pedestrian unless it was near a designated crosswalk.

4.4.3 Requirement 3: Privacy and Data Governance

Privacy and data governance is the most widely recognised and best-regulated dimension of AI ethics. The GDPR has established a baseline in Europe, while the EU’s Data Act (2022) complements the Data Governance Act by specifying who can create value from data and under what circumstances.

Two less-obvious aspects deserve attention. First, **data may contain information we do not expect**. A study using 58,000 volunteers’ Facebook Likes demonstrated that easily accessible digital behavioural records can be used to accurately predict a range of highly sensitive personal attributes: sexual orientation (88% accuracy), political affiliation (85%), ethnicity, religion, intelligence, happiness, substance use, parental separation, age, and gender. The implications – illustrated by the Cambridge Analytica scandal – are profound: consent over primary data does not imply consent over secondary inferences.

Second, **intellectual property rights** raise unresolved questions. Generative AI is trained on real data and produces similar synthetic data, raising the question: who owns the generated output? This applies to music, visual art, poetry, choreography, and film. The debate became vivid when Boris Eldagsen’s AI-generated photograph won a prestigious photography award, prompting Eldagsen himself to decline the prize as a provocation about AI’s role in creative authorship.

4.4.4 Requirement 4: Transparency

Transparency requires ensuring that appropriate information reaches the relevant stakeholders. Humans must be informed of an AI system’s capabilities and limitations, and they

must always be aware when they are interacting with an AI system. Three sub-components are identified: *explanations* (timely and adapted to the audience concerned), *traceability* of AI system behaviour, and *communication* about the system to stakeholders.

The most widely accepted definition of explanation comes from a NIST report: an **explanation** is the evidence, support, or reasoning related to a system’s output or process, where the output of a system differs by task, and the process refers to the procedures, design, and system workflow which underlie the system.

Factors shaping explanations

The form of an appropriate explanation depends on several factors: the type of task for which it is needed; the kind of data on which the AI system operates; who the final user of the explanation is; whether it covers the whole behaviour of the system (*global* explanation) or the reasons for a particular decision instance (*local* explanation); business considerations such as the implications of revealing proprietary model logic; and technical constraints such as whether required information is locally available in a decentralised system.

Bias-in, bias-out

A classic demonstration of why transparency matters is the wolf-vs-husky classification experiment. A system was trained on twenty hand-selected images in which all wolf pictures included snow in the background while husky pictures did not. The model learned to detect snow rather than the animal, predicting “wolf” whenever snow was present and “husky” otherwise. Explanation techniques revealed this spurious cue, which would have remained invisible to accuracy metrics alone.

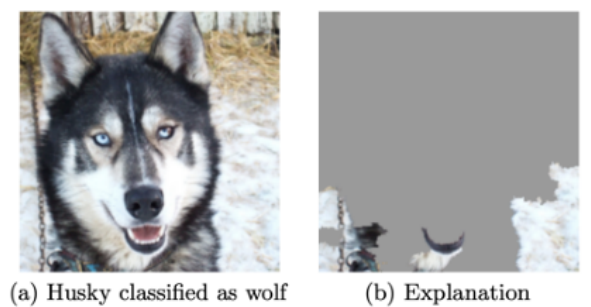


Figure 3: Wolf-vs-husky example: husky image classified as wolf, alongside explanation image highlighting snow background as the decisive feature

4.4.5 Requirement 5: Diversity, Non-Discrimination, and Fairness

This requirement covers three interrelated sub-dimensions: ensuring the inclusion of diverse data and people such that individuals at risk of exclusion have equal access to AI benefits; advocating for **diversity** through heterogeneous and randomised data acquisition procedures and representative population sampling including minorities; and assuring non-discriminating automated processes that do not generate unfair or biased models.

Fairness in EU law

Article 21 of the EU Charter of Fundamental Rights prohibits discrimination on any

ground including sex, race, colour, ethnic or social origin, genetic features, language, religion or belief, political or other opinion, national minority membership, property, birth, disability, age, or sexual orientation. In the AI context, two discrimination scenarios are distinguished:

1. **Direct discrimination** (disparate treatment): an AI system treats individuals differently on the basis of a sensitive attribute.
2. **Indirect discrimination** (disparate impact): a seemingly neutral provision, criterion, or practice disproportionately disadvantages members of a sensitive group relative to others.

Computational fairness

Computational fairness addresses biases and discrimination arising from the use of algorithms, ensuring that systems do not perpetuate or amplify existing societal biases against groups defined by sensitive attributes. **Fairness metrics** are quantitative measurements used to assess and quantify the fairness or bias of an algorithm’s predictions or decisions.

The COMPAS case

The COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) system predicts recidivism and assigns defendants a low, medium, or high risk score. Used in many US states including New York and Wisconsin, COMPAS achieves approximately 70% accuracy. However, an investigation by ProPublica journalists revealed that Black defendants were far more likely than White defendants to be incorrectly classified as high risk, while White defendants were more likely to be incorrectly flagged as low risk – a paradigmatic example of disparate impact arising from racially skewed historical data.

Amazon’s recruiting tool

Amazon shut down an experimental AI recruiting tool after discovering it discriminated against women. The tool was designed to crawl the web, identify potential candidates, and rate them on a one-to-five star scale. The algorithm systematically downgraded women’s CVs for technical roles such as software developer because it had been trained on ten years of historical recruitment data that was dominated by male profiles, encoding historical underrepresentation as a predictive signal.

4.4.6 Requirement 6: Societal and Environmental Wellbeing

AI-based systems should benefit all of humankind, not only at the present time but also for future generations. Accordingly, AI systems must be **sustainable and environmentally friendly**, such that their adoption does not entail progressive depletion of natural resources or disruption of ecological balance.

In practice AI requires high-performance computing with a heavy carbon footprint. Paradoxically, AI can also help reduce that same footprint through power-aware workload management, optimal allocation and scheduling of computing resources, cooling optimisation and thermal-aware workload dispatching, and anomaly detection in energy systems. Today’s supercomputers are power-limited and the acceptable power budget for an Exascale machine is approximately 20 MW; power capping – constraining consumption within a defined budget – is a widespread management strategy.

At the societal level, AI can improve welfare by performing routine tasks more safely and efficiently, enhancing productivity and quality of life, streamlining administrative processes, and helping city planners visualise the consequences of climate change or predict future floods.

4.4.7 Requirement 7: Accountability

Accountability means ensuring responsibility for the development, deployment, maintenance, and use of AI systems and their outcomes. When an AI system causes harm, it is necessary to identify who is responsible – the algorithm’s author, the owner of the object in which the algorithm operates, or the supervising human – and legal frameworks must assign that responsibility clearly.

Auditability

A key mechanism for realising accountability is **auditability**: the development of practical tools capable of verifying desirable properties of neural networks such as stability, sensitivity, relevance, and reachability, as well as metrics beyond explainability such as traceability, data quality, and integrity. Standards bodies including IEEE, ISO/IEC, and CEN/CENELEC are materialising concrete guidelines to apply trustworthy AI requirements in industrial settings.

4.5 AI for Social Good

AI for Social Good (AI4SG) refers to the use of AI to reduce, mitigate, or eradicate social or environmental problems without introducing new harms or amplifying existing ones. The United Nations Sustainable Development Goals provide a natural framework: the seventeen SDGs – covering poverty elimination, zero hunger, health and well-being, quality education, gender equality, clean water, affordable energy, decent work, industry and infrastructure, reduced inequalities, sustainable cities, responsible consumption, climate action, life below water, life on land, peace and justice, and partnerships – represent domains in which AI applications can create genuine social value when designed with ethical rigour.

4.6 Where Does AI Ethics Enter the ML Pipeline?

A core message of the computational perspective on AI ethics is that ethical requirements are *not* external to AI and machine learning. They are embedded throughout the entire lifecycle of an AI system. Every stage of the machine learning pipeline is touched by at least one ethical dimension:

- **Data collection** – bias and representativeness
- **Data preprocessing** – privacy and data governance
- **Feature engineering** – fairness constraints
- **Model selection** – robustness and safety
- **Training** – explainability
- **Evaluation** – accountability
- **Deployment** – auditability
- **Monitoring** – all of the above, continuously

Subsequent lectures will study practical computational approaches to these dimensions: where they plug into the pipeline, how they modify learning objectives or constraints, and how trade-offs are formally managed.

4.7 Current Approaches: Ethics as a Maturity Model

Organisations do not adopt ethical AI instantaneously. They typically evolve along a maturity trajectory from *awareness* towards *ethical-by-design*, passing through compliance and governance layers. The table below summarises this progression.

Level	Focus	Practices / Artefacts	Limit
Awareness	Recognition of impacts	Ethical principles, guidelines, training	Low operationalisation; risk of ethics washing
Compliance	Regulatory baseline	Risk classification, documentation, model cards, datasheets	Reactive; checks often late in lifecycle
Governance	Organisational control	Roles (RACI), audit trails, internal review boards, monitoring	Controls may not be integrated into MLOps
Ethical-by-design	Engineering integration	Ethical constraints as requirements; testing gates; explicit trade-offs	Requires metrics, automation, continuous monitoring

Table 2: A maturity model for ethics and responsible AI (adapted from HLEG 2019, OECD 2019, NIST 2023, ISO/IEC 2023).

More is not always better

A tempting but mistaken conclusion is that ethical AI simply requires more: more data, more algorithms, more computational power. The dataset usage map across machine learning research reveals a stark geographic skew – datasets originating in the United States account for the largest share of usage across 26,335 research papers from 2015 to 2020 – meaning that AI as currently practised sees and models a decidedly non-representative slice of the world.

Algorithmically, AI is a *rational* system: AI agents have preferences or priorities over outcomes and optimise actions based on those preferences. The dominant paradigm thus encodes assumptions – optimisation, efficiency, rationality, agency, autonomy – that reflect what one might call **Global North stereotypes**. Human behaviour, by contrast, is deeply contextual: people pursue seemingly incompatible goals, hold inconsistent beliefs, and are motivated by altruism, fairness, justice, and the desire to avoid future regret. Actions are shaped by context, by others, and by the particulars of situations in ways that purely rational optimisation frameworks struggle to capture.

The central challenge, then, is not just engineering competence but moral and political awareness: **impact means human responsibility**. Decisions about what to optimise, whose preferences to weight, which trade-offs to accept, and which harms to prevent are irreducibly ethical decisions, not merely technical ones.

5 L2 – Diversity, Non-Discrimination and Fairness

5.1 Why Fairness?

Society is facing a dramatic increase in **pervasive inequality** and **intersectional discrimination** due to the widespread use of AI [?, ?]. Machine learning is actively contributing to creating a society where some groups or individuals are systematically disadvantaged. Real-world incidents – from racially biased recidivism scoring tools to gender-biased hiring algorithms and discriminatory credit systems – document how both traditional and generative AI models reproduce and amplify social biases. Image-generation models (e.g., Stable Diffusion) reproduce occupational stereotypes: queries for “judge” or “doctor” yield predominantly light-skinned male faces, while “dishwasher worker” or “cashier” skew towards darker skin tones and female faces respectively. Language models are not exempt: GPT-3.5 gave opposite moral verdicts on structurally identical scenarios depending solely on whether the retaliating nation was described as Jewish or Muslim, illustrating how bias can manifest as **inconsistent moral reasoning**. Attempts at debiasing have introduced new problems, as seen when Google’s image generator overcorrected by injecting demographic diversity into historically homogeneous groups.

5.2 Fairness and Bias in AI

What is Fairness?

Article 21 of the EU Charter of Fundamental Rights prohibits discrimination on grounds including sex, race, colour, ethnic or social origin, genetic features, language, religion or belief, political opinion, membership of a national minority, property, birth, disability, age, and sexual orientation. Two distinct discrimination scenarios are distinguished. **Direct discrimination** (disparate *treatment*) occurs when a person is treated less favourably because of a protected attribute. **Indirect discrimination** (disparate *impact*) arises when a seemingly neutral provision, criterion, or practice disproportionately disadvantages members of a given sensitive group compared to others.

What is Bias?

Fairness and **bias** are two sides of the same coin. While no universally agreed definition of fairness exists, it can be broadly understood as the absence of prejudice or preference for an individual or group based on their characteristics – that is, the absence of bias. **Bias in AI** is the phenomenon that occurs when an AI system produces results that are systemically prejudiced. Bias takes many shapes and forms and can be introduced at any stage of the model development pipeline: from the world to the data (historical bias, representation bias, measurement bias, temporal bias, omitted variable bias), within the AI/ML model itself (algorithmic bias, evaluation bias, aggregation bias, popularity bias, ranking bias, emergent bias, linking bias), and in the human-review and action stages (behavioural bias, presentation bias, content-production bias, social bias). **Algorithmic bias** in particular can stem from: inadvertent privacy violations; programmer-imposed priorities and hierarchies for data assessment; data collection based on human-selected criteria that reflects designer bias; and reinforcement of stereotypes through relevance filtering based on prior user behaviour.

Computational Fairness

Computational fairness addresses the potential biases and discrimination that can

arise from the use of computational algorithms, with the goal of ensuring that algorithms do not perpetuate or amplify existing biases, and do not discriminate against certain groups based on **sensitive attributes**. A key instrument for this is the **fairness metric**: a quantitative measurement used to assess and quantify the fairness or bias of an algorithm’s predictions or decisions.

5.3 Fairness Awareness

Operationalising fairness requires two elements: (1) a *definition of fairness notions*, which is context-dependent and informed by social perspectives; and (2) a *quantitative mechanism* to measure them. Most approaches are built on the notion of **protected** (or **sensitive**) **variables** and **(un)privileged groups**. Groups – defined by one or more sensitive variables – are those that are disproportionately (less) more likely to receive a positive classification. Protected variables define the aspects of data that are socioculturally precarious for the application of ML: gender, ethnicity, age, their synonyms, and essentially any other feature of the data that involves or concerns people.

5.4 Fairness Awareness in ML

Procedural fairness	Outcome fairness		
		Observational	Causal
1) Fairness through unawareness	Group	2) Independence 3) Separation 4) Sufficiency	5) Causality
	Individual	6) Individual fairness	

Figure 4: Organising framework of algorithmic fairness metrics: two-dimensional table with Procedural vs Outcome fairness (columns) and Group vs Individual granularity (rows), with observational and causal sub-columns

The landscape of fairness metrics is organised along two orthogonal axes, giving rise to a taxonomy of six main families.

Procedural Fairness

Procedural fairness is a concept inherited from administrative law, concerned with equality of treatment *within the process* that carries out a decision. In the computational domain it corresponds to **fairness through unawareness**: simply omitting sensitive attributes from the algorithm. This approach is widely recognised as ineffective [?]: (i) model accuracy tends to decrease; (ii) discrimination effects do not improve because removing a sensitive variable does not sever its relationships with proxy variables; (iii) bias may be caused by a combination of variables rather than a single one. Paradoxically, omissions can therefore *increase* bias or discrimination [?].

Outcome Fairness

Outcome fairness targets equality of the outputs (a *fair result*) and splits into two orthogonal dimensions. The first distinguishes **individual** notions – which compare outcomes for single individuals – from **group** notions – which aggregate outcomes across all individuals belonging to the same sensitive category. The second dimension distinguishes **observational** approaches, which rely on joint distributions of observable quantities (outcomes, decisions, features, sensitive attributes), from **causal** approaches, which require causal inference to uncover the relations between variables.

5.4.1 Observational Fairness: the Three Group Criteria

Group-level observational fairness metrics are built upon three abstract criteria, expressed in terms of a classifier’s sensitive variable A , target variable Y , and classification score (or decision) R/D . A fundamental result is that these three criteria are **mutually exclusive** when base rates differ across groups and the classifier is imperfect.

Independence

Independence ensures that the equality of outcomes is independent of the sensitive attribute; formally, $A \perp D$. In binary classification this reduces to requiring the same acceptance rate across all groups:

$$P\{D = 1 \mid A = a\} = P\{D = 1 \mid A = b\}.$$

A **relaxed** version introduces a slack $\varepsilon > 0$, either as a difference condition $P\{D = 1 \mid A = a\} \geq P\{D = 1 \mid A = b\} - \varepsilon$ or a ratio condition $P\{D = 1 \mid A = a\}/P\{D = 1 \mid A = b\} \geq 1 - \varepsilon$. Named metrics equivalent or related to independence include *statistical parity*, *group fairness*, *demographic parity*, and the *Darlington criterion* (4).

Independence has a crucial **limitation**: it ignores any genuine correlation between A and Y . Satisfying equal acceptance rates does not preclude assigning the positive label at random within each group, so two similarly qualified individuals from different groups may be treated very differently – independence does not guarantee individual fairness [?].

Separation

Separation captures “similar people should be treated similarly” as *equality of errors*: formally, $D \perp A \mid Y$. In binary classification this requires, for all groups a, b :

$$P\{D = 0 \mid Y = 1, A = a\} = P\{D = 0 \mid Y = 1, A = b\},$$

$$P\{D = 1 \mid Y = 0, A = a\} = P\{D = 1 \mid Y = 0, A = b\}.$$

In the hiring example, separation demands equal false-negative rates (similar rejection rates for equally qualified candidates) and equal false-positive rates across genders. Named metrics include *equalized odds* (equivalent), *equal opportunity* (relaxation), *conditional procedure accuracy*, and *avoiding disparate mistreatment*. Separation may be preferable to independence when a genuine correlation between the sensitive feature and the outcome exists; its limitations are mainly those inherent to any observational metric.

Sufficiency (Calibration)

Sufficiency ensures that the model’s score R is informative enough that knowing A adds nothing further; formally, $Y \perp A \mid R$. For all groups a, b and all values r :

$$P\{Y = 1 \mid R = r, A = a\} = P\{Y = 1 \mid R = r, A = b\}.$$

This means that among all individuals receiving a predicted probability p , the same fraction p should actually be positive instances regardless of group membership – i.e., the chances of being truly qualified given a positive hiring decision are the same for men and women. Named metrics include the *Cleary model*, *conditional use accuracy*, and *predictive parity*. Calibration is necessary but not sufficient for fairness; its limitations are again mainly observational.

5.4.2 Observational Fairness: Advantages and Limitations

Observational metrics share several advantages: they are easy to state, require only a lightweight formalism, and impose no assumptions about the inner workings of the classifier, the causal impact of decisions, or the correlations between features and outcomes. Their major drawback is precisely that they evaluate only what is observable; hidden relationships, unobservable confounders, and the downstream consequences of decisions lie entirely outside their scope [?].

5.4.3 Causal Fairness

Causal fairness exploits the causal graph together with observed data to discover hidden relationships and identify discrimination at its root causes rather than merely on observed disparities.

The most prominent causal notion is **counterfactual fairness** [?]. Given a causal model (U, V, F) – where U are latent background variables, $V = S \cup X$ are observable variables including the sensitive variable S , and F defines structural equations – a decision-making process is counterfactually fair if, for any individual, the outcome distribution is identical in the actual world and in any counterfactual world in which the individual’s sensitive attributes are changed:

$$P(\hat{y}_{A \leftarrow a}(U) = y \mid X = x, A = a) = P(\hat{y}_{A \leftarrow a'}(U) = y \mid X = x, A = a).$$

In the hiring example one could discover that the outcome Y is driven by educational achievement E and interview score I , both of which are themselves causally affected by the protected attribute A (gender); the causal approach targets this root-cause path rather than surface-level outcome disparities.

Advantages of causal fairness include deeper understanding enabling more effective interventions, more precise fairness measures, and the possibility of *fairness by design*. Limitations include the complexity of constructing accurate causal graphs, requirements for domain expertise and high-quality data, computational resource intensity, reduced interpretability compared to traditional ML models, and potential conflicts with legal and regulatory requirements.

5.4.4 Individual Fairness

Individual fairness criteria follow the principle that *similar individuals should receive similar treatments*, working on a notion of similarity rather than group membership. Two approaches exist. **Fairness through unawareness** (discussed above) is the naive version. **Fairness through awareness** [?] stresses the importance of choosing an appropriate target-related distance metric to assess which individuals are similar in a given context, expressed formally as a Lipschitz condition on the mapping from individuals to distributions over outcomes. The similarity metric is in practice difficult to define and the condition is computationally infeasible at scale.

5.4.5 The Impossibility of Simultaneous Fairness Criteria

A fundamental result in algorithmic fairness states that, when base rates differ across groups (i.e., $P(Y = 1 \mid A = a) \neq P(Y = 1 \mid A = b)$) and the classifier is imperfect, it is **impossible** to simultaneously satisfy independence, separation, and sufficiency [?, ?]. Fairness metrics are therefore not merely different – they are **structurally incompatible** in most realistic settings. Practitioners must choose which criterion best matches the ethical and legal context of their application.

5.5 Fairness Awareness in Generative AI

Overview

Fairness evaluation for large generative models (LLMs) targets **representations, probabilities, and generated text** rather than only classification outcomes, and assessment must combine automatic metrics with human judgment. Three complementary perspectives structure the evaluation: **embedding-based** approaches measure bias in vector representations (word or sentence embeddings); **probability-based** approaches inspect token or sequence probabilities under controlled prompt perturbations; and **generated-text-based** approaches analyse actual model outputs for distributional or semantic biases such as toxicity, stereotyping, and demographic representation rates.

Metric	Equation	Reference
EMBEDDING-BASED		
WORD EMBEDDING		
WEAT	$f(A, W) = \frac{\text{mean}_{a_1 \in A_1} s(a_1, W_1, W_2) - \text{mean}_{a_2 \in A_2} s(a_2, W_1, W_2)}{\text{std}_{a \in A} s(a, W_1, W_2)}$	(Caliskan, Bryson, & Narayanan, 2017)
SENTENCE EMBEDDING		
SEAT	$f(S_A, S_W) = \text{WEAT}(S_A, S_W)$	(May, Wang, Bordia, Bowman, & Rudinger, 2019)
CEAT	$f(S_A, S_W) = \frac{\sum_{i=1}^N v_i \text{WEAT}(S_{A_i}, S_{W_i})}{\sum_{i=1}^N v_i}$	(Guo & Caliskan, 2021)
Sentence Bias Score	$f(S) = \sum_{s \in S} \cos(s, v_{\text{gender}}) \cdot \alpha_s $	(Dolci, Azzalini, & Tanelli, 2023)
PROBABILITY-BASED		
MASKED TOKEN		
DisCo	$f(S) = \mathbb{I}(\hat{y}_{i, [MASK]} = \hat{y}_{j, [MASK]})$	(Webster et al., 2021)
Log-Probability Bias Score	$f(S) = \log \frac{p_{a_i}}{p_{\text{prior}_i}} - \log \frac{p_{a_j}}{p_{\text{prior}_j}}$	(Kurita, Vyas, Pareek, Black, & Tsvetkov, 2019)
Categorical Bias Score	$f(S) = \frac{1}{ W_1 } \sum_{w \in W} \text{Var}_{a \in A} \log \frac{p_a}{p_{\text{prior}}}$	(Ahn & Oh, 2021)
PSEUDO-LOG-LIKELIHOOD		
CrowS-Pairs Score	$f(S) = \sum_{u \in U} \log P(u U_{\setminus u}, M; \theta)$	(Nangia, Vania, Bhalerao, & Bowman, 2020)
Context Association Test	$f(S) = \frac{1}{ M } \sum_{m \in M} \log P(m U; \theta)$	(Nadeem, Bethke, & Reddy, 2021)
All Unmasked Likelihood	$f(S) = \frac{1}{ S_1 } \sum_{s \in S} \log P(s S; \theta)$	(Kaneko & Bollegala, 2022)
Language Model Bias	$f(S) = t\text{-value}(PP(S_1), PP(S_2))$	(Barikeri, Lauscher, Vulić, & Glavaš, 2021)
GENERATED TEXT-BASED		
DISTRIBUTION		
Social Group Substitution	$f(\hat{Y}) = \psi(\hat{Y}_i, \hat{Y}_j)$	(Gallegos et al., 2024)
Co-Occurrence Bias Score	$f(w) = \log \frac{P(w A_i)}{P(w A_j)}$	(Bordia & Bowman, 2019)
Demographic Representation	$f(G) = \sum_{a \in A} \sum_{\hat{Y} \in \hat{\mathcal{Y}}} C(a, \hat{Y})$	(Liang et al., 2023)
Stereotypical Associations	$f(w) = \sum_{a \in A} \sum_{\hat{Y} \in \hat{\mathcal{Y}}} C(a, \hat{Y}) \mathbb{I}(C(w, \hat{Y}) > 0)$	(Liang et al., 2023)
CLASSIFIER		
Perspective API	$f(\hat{Y}) = c(\hat{Y})$	
Expected Maximum Toxicity	$f(\hat{Y}) = \max_{\hat{Y} \in \hat{\mathcal{Y}}} c(\hat{Y})$	(Gehman, Gururangan, Sap, Choi, & Smith, 2020)
(Empirical) Toxicity Probability	$f(\hat{Y}) = P\left(\sum_{\hat{Y} \in \hat{\mathcal{Y}}} \mathbb{I}(c(\hat{Y}) \geq 0.5) \geq 1\right)$	(Gehman et al., 2020)
Toxicity Fraction	$f(\hat{Y}) = \mathbb{E}_{\hat{Y} \in \hat{\mathcal{Y}}} [\mathbb{I}(c(\hat{Y}) \geq 0.5)]$	(Liang et al., 2023)
Score Parity	$f(\hat{Y}) = \mathbb{E}_{\hat{Y} \in \hat{\mathcal{Y}}} [c(\hat{Y}_i, i) A = i] - \mathbb{E}_{\hat{Y} \in \hat{\mathcal{Y}}} [c(\hat{Y}_j, j) A = j] $	(Sicilia & Alikhani, 2023)
Counterfactual Sentiment Bias	$f(\hat{Y}) = W_1 \left(P(c(\hat{Y}_i) A = i), P(c(\hat{Y}_j) A = j) \right)$ (P.-S. Huang et al., 2020)	
Regard Score	$f(\hat{Y}) = c(\hat{Y})$	(Sheng, Chang, Natarajan, & Peng, 2019)
Full Gen Bias	$f(\hat{Y}) = \sum_{i=1}^C \text{Var}_{w \in W} \left(\frac{1}{ \hat{\mathcal{Y}}_w } \sum_{\hat{Y}_w \in \hat{\mathcal{Y}}_w} c(\hat{Y}_w)[i] \right)$	(Smith, Hall, Kambadur, Pre-sani, & Williams, 2022)
LEXICON		
HONEST	$f(\hat{Y}) = \frac{1}{ \hat{\mathcal{Y}} \cdot k} \sum_{\hat{Y}^k \in \hat{\mathcal{Y}}^k} \sum_{\hat{g} \in \hat{\mathcal{Y}}^k} I_{\text{HurtLex}}(\hat{g})$	(Nozza, Bianchi, & Hovy, 2021)
Psycholinguistic Norms	$f(\hat{Y}) = \frac{\sum_{\hat{Y} \in \hat{\mathcal{Y}}} \sum_{\hat{g} \in \hat{\mathcal{Y}}} \text{sign}(\text{affect}(\hat{g})) \cdot \text{affect}(\hat{g})^2}{\sum_{\hat{Y} \in \hat{\mathcal{Y}}} \sum_{\hat{g} \in \hat{\mathcal{Y}}} \text{affect}(\hat{g}) }$	(Dhamala et al., 2021)
Gender Polarity	$f(\hat{Y}) = \frac{\sum_{\hat{Y} \in \hat{\mathcal{Y}}} \sum_{\hat{g} \in \hat{\mathcal{Y}}} \text{sign}(\text{bias}(\hat{g})) \cdot \text{bias}(\hat{g})^2}{\sum_{\hat{Y} \in \hat{\mathcal{Y}}} \sum_{\hat{g} \in \hat{\mathcal{Y}}} \text{bias}(\hat{g}) }$	(Dhamala et al., 2021)

Figure 5: Taxonomy of Evaluation Metrics for LLMs

What Changes Compared to Fairness in ML?

Several fundamental differences distinguish the generative AI setting from the classical ML one. Regarding the **object of analysis**, ML targets discrete decision labels, while LLMs produce representations, conditional probabilities, and open-ended text. Regarding **model access**, ML systems often expose full prediction scores; LLMs may be black boxes accessible only via sampling or API calls, rendering some metrics infeasible. LLMs also exhibit **multiplicity of outputs**: because they produce distributions over continuations, fairness must be defined over a *set* of sampled outputs rather than a single prediction. **Prompt sensitivity** is an additional complication: even minor prompt changes can induce large demographic differences, making prompt engineering a first-class fairness concern. **Evaluation noise** arising from sampling stochasticity requires larger sample

sizes and explicit uncertainty quantification; fairness metrics are estimates of stochastic outputs, not deterministic properties of a fixed classifier, so statistical confidence intervals must always be reported. Finally, generative models can cause **representational harms** – amplifying stereotypes or under-/over-representing groups in text – a harm class distinct from the disparate treatment studied in classification.

Practical Workflow

A principled fairness evaluation in GenAI proceeds in four stages. **Scope and requirements:** define the harms of interest (representation, toxicity, stereotyping, omission) and choose protected attributes with operational definitions. **Metric selection:** use a multi-pronged combination of embedding-based and generated-text-based metrics for robustness; when only API access is available, prefer generated-text metrics and probability-based proxies. **Prompt design and sampling:** employ systematic prompt templates and controlled counterfactual perturbations; draw sufficiently large sets of continuations and report confidence intervals. **Reporting and remediation:** document the metric suite, sampling details, and limitations; choose remediation strategies aligned with the identified harm, which may include prompt filters, decoding constraints, data balancing, fine-tuning, or safety classifiers. Common pitfalls are over-reliance on a single metric, small or non-representative prompt sets, and neglecting intersectional and multilingual issues.

5.6 Fairness Awareness: Challenges

Which Sensitive Attributes?

A first open challenge concerns the choice of attributes to protect: which variables should be treated as protected, which variables are correlated with or act as proxies for those attributes, and which combinations of variables act as **quasi-identifiers** that permit indirect identification when combined?

Which Notions of Fairness?

As established by the impossibility results, fairness metrics are often conflicting and involve trade-offs with accuracy and related performance metrics. Desirable theoretical properties – such as an **incrementally conservative fairness measure** (satisfaction of the metric does not decrease as predictor accuracy increases) or a **dataset metric** (the measure can be computed from data alone) – remain research goals rather than fully realised tools.

The EU Gold Standard

The EU *AI Act* (14 June 2023) addresses fairness primarily through Article 4a(1)(e) – which requires AI systems to be developed in ways that include diverse actors, promote equal access and gender equality, and avoid discriminatory impacts and unfair biases prohibited by Union or national law – and Article 10(2)(f), which mandates examination of possible biases that could negatively impact fundamental rights. Article 28b(b) further requires foundation models to incorporate only datasets subject to appropriate governance measures, including examination for possible biases and appropriate mitigation.

Despite these provisions, a set of persistent challenges remain (summarised from Kattinig et al., *Assessing Trustworthy AI*):

- **Challenge 1:** Fairness is not defined precisely or uniformly in the Act. It is ambiguous when fairness is actually satisfied or an unfair bias is to be assumed; concrete compliance requirements and criteria are missing.
- **Challenge 2:** No guidance exists on which measure(s) should be used for different contexts.
- **Challenge 3:** No guidance on how to handle conflicts among various fairness measures and between accuracy and fairness.
- **Challenge 4:** Fairness thresholds (e.g., the ε in the 4/5-ths rule for disparate impact) are not determined or guided by the legislation.
- **Challenge 5:** Many metrics and mitigation methods have been developed, but few meet legal requirements. A clear gap exists between the statistical measures embedded in fairness toolkits and the context-sensitive, often intuitive evidential standards used by the European Court of Justice (Wachter et al., *Why Fairness Cannot Be Automated*).
- **Challenge 7:** Detection, prevention, and mitigation of unfairness for multi-modal generative AI is substantially harder than for unimodal classification.

5.7 Towards Structured Knowledge and Ontologies

The preceding sections have introduced multiple bias types, multiple fairness notions, multiple metrics, and (as will be seen) multiple intervention strategies. The key challenge is that these concepts do not exist in isolation: one needs to know which metric corresponds to which notion, and which notion addresses which type of bias. This calls for a **structured conceptual model**.

Fairness knowledge can be organised into three conceptual layers. The **social layer** encompasses social principles, protected groups, and types of discrimination. The **algorithmic layer** encompasses bias types and fairness notions. The **operational layer** encompasses fairness metrics, evaluation procedures, and mitigation techniques. A key insight follows: *fairness is not a number – it is a structured socio-technical relation*, captured by the pipeline Bias $\rightarrow$ Notion $\rightarrow$ Metric $\rightarrow$ Measurement $\rightarrow$ Action.

Knowledge Representation Paradigms

Fairness knowledge can be represented at different levels of formality: **narrative and conceptual frameworks** (ethical principles, legal doctrines such as the EU AI Act and GDPR); **taxonomies and structured vocabularies** (bias types, fairness notions, metric families); **formal ontologies** in OWL/RDF, providing machine-readable classes and relations that support logical reasoning and interoperability; and **executable specifications** such as rules, constraints, and compliance checks that can be integrated directly into ML pipelines. The course’s focus is on the last category: *computational ontologies enabling operational fairness*.

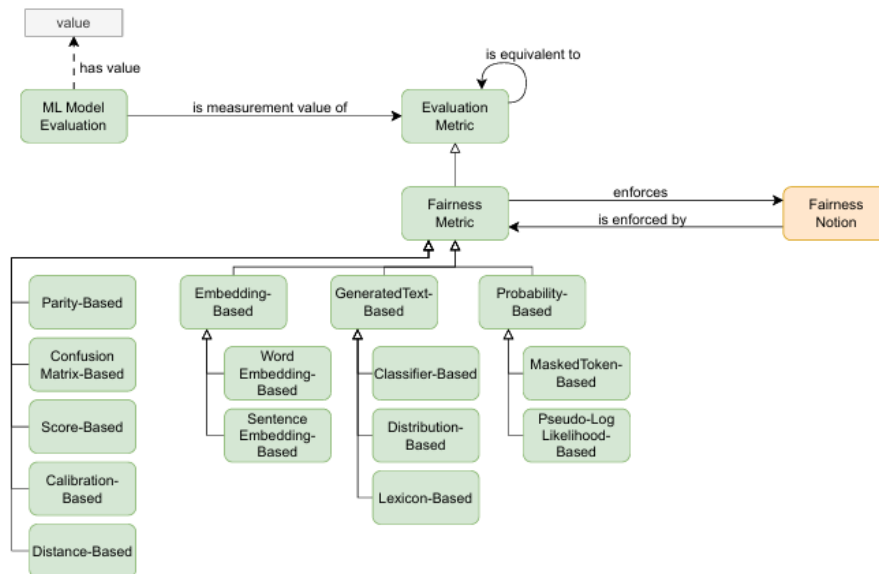
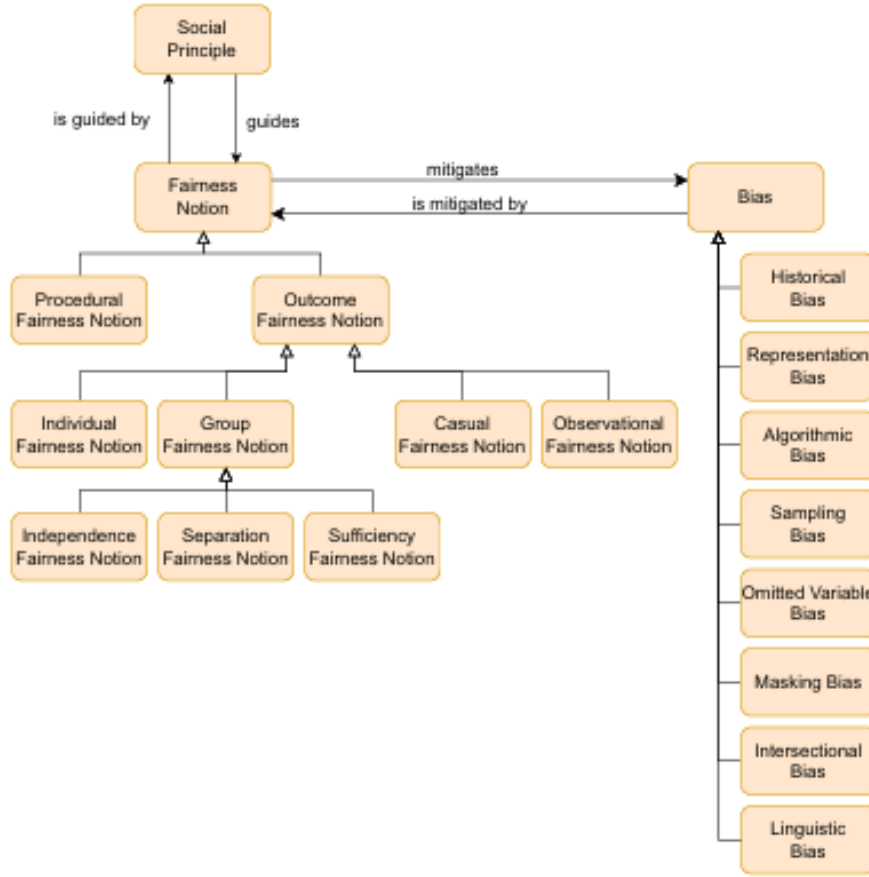
Why Ontologies?

Formal ontologies offer several advantages for operationalising fairness. They provide **conceptual clarity**, disambiguating bias types, fairness notions, and metrics. They enable **interoperability** by linking fairness, risk, and privacy vocabularies and supporting cross-domain reasoning to bridge the legal–technical divide. They offer **reasoning support**, allowing a system to suggest appropriate fairness metrics given a context, and to detect inconsistencies or missing elements in a fairness evaluation. Finally, they enable

traceability and auditability by connecting bias source to fairness notion, metric, and mitigation strategy along the entire development lifecycle.

Computational Ontologies for AI Fairness

Several ontologies have been proposed in the literature for different aspects of AI fairness.



Ontology	Primary Focus	Contribution	Orientation
BODP	Abstract bias modelling	Formal pattern for bias structure	Conceptual
FMO	Fairness metrics	Formalises definitions, metric selection	Evaluation-centric
AIRO/VAIR	AI risk representation	High-risk systems, EU AI Act alignment	Compliance
DPV	Data protection	W3C vocabulary for fair data processing	Privacy/governance
Doc-Bias	Bias provenance	Neuro-symbolic bias modelling across ML lifecycle	Hybrid/lifecycle
Croissant ACROCPoLis	Dataset metadata	Structured metadata and stakeholder perspectives	Complementary

Computational ontologies for AI fairness (adapted from the lecture slides).

6 Diversity, Non-Discrimination and Fairness — Mitigation

6.1 From Fairness Awareness to Mitigation

Fairness work in machine learning proceeds in two complementary stages. **Awareness** answers the diagnostic questions: is the system unfair, and according to which notion? **Mitigation** then answers the prescriptive questions: how can we modify the system to reduce unfairness, at what cost, and under which constraints? Formally, mitigation is an *intervention on the learning process or its outputs* — it is not merely a measurement activity but an active reshaping of the pipeline.

Mitigation techniques can alter several different components of a machine learning system: the **data distribution**, the **learning objective**, the **decision boundary**, the **representation space**, or the **output calibration**. The key insight is that every mitigation strategy ultimately reshapes one or more of three probabilistic objects: $P(X)$, $P(Y | X)$, or $P(\hat{Y} | X)$.

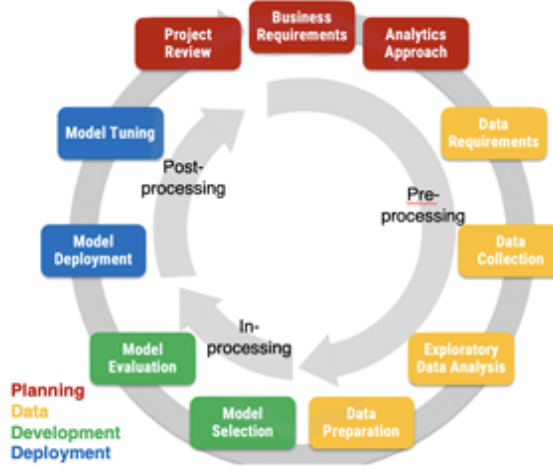
This flexibility comes with structural trade-offs. Imposing fairness constraints may reduce predictive accuracy; different fairness notions lead to different, sometimes incompatible, mitigation strategies; and interventions may themselves introduce new biases. Two fundamental questions therefore precede any mitigation effort: which notion of fairness is being enforced (the awareness question), and where in the pipeline can we intervene?

6.2 Where Can We Act? Timing of Intervention

Fairness interventions can be placed at three distinct stages of the AI lifecycle.

Pre-processing

Pre-processing methods modify the **training data before learning begins**. Because the data is the most upstream component of the pipeline, this stage is argued to be the most flexible point for repairing bias: any downstream model can subsequently



be trained on the corrected data without modification, making the approach **model-agnostic**. The main techniques include **reweighing**, **resampling**, **relabelling**, the **Disparate Impact Remover**, and **counterfactual data augmentation**. The principal limitation is that modifying the data may distort the true underlying distribution, and certain transformations may conflict with data-protection regulations such as the GDPR, potentially introducing new biases in the process.

In-processing

In-processing methods **modify the learning algorithm itself**, so that fairness is enforced during model training. The standard training objective $\min_{\theta} \mathcal{L}(\theta)$ — which optimises only prediction accuracy — is augmented with a fairness penalty:

$$\min_{\theta} \mathcal{L}(\theta) + \lambda \Phi_{\text{fair}}(\theta)$$

where $\Phi_{\text{fair}}(\theta)$ is a differentiable fairness penalty term and $\lambda \geq 0$ controls the fairness–accuracy trade-off. Concrete in-processing families include **regularisation-based methods**, **constrained optimisation**, **adversarial debiasing**, and **fair representation learning**. The advantages are direct control over the trade-off, often stronger fairness guarantees, and the ability to target specific fairness notions explicitly. The limitations are that the approach requires access to the training procedure, incurs higher computational complexity, may reduce model interpretability, and is sensitive to the choice of the hyperparameter λ .

Post-processing

Post-processing methods **modify the model outputs or decision rule without retraining**. Representative techniques include **threshold adjustment per group**, **equalized odds post-processing**, **calibration adjustment**, and **rejection option classification**. Because these methods operate on the outputs of a frozen model, they do not require access to training data and are applicable even to black-box models, making them simple to implement. Their key limitation is that accuracy may be significantly reduced, and they adjust decisions without correcting biased features or latent representation spaces.

6.3 Enforcing Fairness in ML: Detailed Techniques

6.3.1 Taxonomy of Intervention Techniques

The table above organises the literature by two dimensions: the *stage of intervention* (rows) and the *fairness notion being enforced* (columns). Procedural fairness (fairness through unawareness) sits alongside group fairness notions — independence/demographic parity, separation/equalized odds, sufficiency, and causal fairness — and individual fairness notions. This mapping makes explicit that the choice of fairness criterion constrains which mitigation techniques are applicable.

6.3.2 Pre-processing: Disparate Impact Remover

Idea and core mechanism

The **Disparate Impact Remover (DIR)**, introduced by Feldman et al. (2015), modifies feature values to remove the statistical dependence between the protected attribute A and the input features X , while preserving the relative ordering of individuals *within* each protected group (rank-preserving transformation). For each feature X and each group $A = a$, the algorithm computes the empirical group-conditional cumulative distribution function $F_{X|A=a}$, maps each value to its within-group percentile (its *rank*), and then replaces it with the value at the same percentile in a common reference (pooled) distribution:

$$X' = F_{\text{target}}^{-1}(F_{X|A=a}(X)).$$

The net effect is that after transformation, $P(X' | A = a) \approx P(X' | A = b)$ for any two groups a and b , thereby reducing disparate impact before any model is trained.

The repair level parameter

The transformation is parameterised by a **repair_level** $\lambda \in [0, 1]$, which interpolates smoothly between the original and the fully repaired feature:

$$X' = (1 - \lambda) X + \lambda X_{\text{repaired}}.$$

At $\lambda = 0$ the data are unchanged; at $\lambda = 1$ the distributions are fully aligned across groups; at intermediate values, a partial alignment is achieved.

Step-by-step toy example

Consider a dataset with four individuals characterised by sex (A), income (X), and a binary label (Y):

ID	$A = \text{Sex}$	$X = \text{Income}$	$Y = \text{Label}$
1	Male	70	1
2	Male	60	1
3	Female	40	0
4	Female	30	0

Step 1 — within-group percentiles. Each individual is assigned their percentile rank within their protected group: individual 1 is at the 100th percentile of the male distribution, individual 2 at the 50th, individual 3 at the 75th of the female distribution, and individual 4 at the 25th.

Step 2 — pooled reference quantiles. The pooled distribution over all four incomes $\{30, 40, 60, 70\}$ yields approximate quantile values: 35 at the 25th percentile, 50 at the 50th, 65 at the 75th, and 70 at the 100th.

Step 3 — mapping. Each individual’s original value is replaced by the pooled quantile at their within-group rank:

ID	Sex (A)	X (orig)	Rank $r = F_{X A}(X)$	$X' = F_{\text{pooled}}^{-1}(r)$
1	Male	70	1.00	70
2	Male	60	0.50	50
3	Female	40	0.75	65
4	Female	30	0.25	35

The resulting dataset has transformed incomes $\{70, 50, 65, 35\}$. Crucially, *only feature values changed*: protected group membership, labels, and the number of rows are all unmodified.

What DIR does and does not guarantee

DIR reduces the observed dependence between X and A — that is, it makes $P(X' | A)$ more similar across groups — but it does not by itself guarantee other fairness criteria such as equalized odds or causal fairness. The magnitude of the feature shift is proportional to how different the group-conditional CDF is from the pooled CDF. The method is model-agnostic but carries two principal limitations: it may remove genuinely predictive signal from the features, and it addresses only observational dependence, not individual fairness.

AIF360 implementation

In the IBM AIF360 library, the pattern is: wrap a dataframe in `StandardDataset` (specifying the label, the protected attribute, and the privileged class), instantiate `DisparateImpactRemover(repair_level=1.0)`, call `fit_transform`, and then train any downstream classifier on the repaired features.

6.3.3 Pre-processing: Reweighting

Idea

Reweighting, proposed by Kamiran and Calders (2010), leaves feature values and labels entirely unchanged. Instead, it assigns a scalar sample weight $W(x)$ to each training observation so that, in the weighted dataset, the sensitive attribute A and the outcome Y are *statistically independent*. The weight for an observation with group $A = a$ and label $Y = y$ is:

$$W(x) = \frac{P(A = a) \cdot P(Y = y)}{P(A = a, Y = y)}.$$

If a group–label combination is under-represented relative to independence, $W > 1$; if over-represented, $W < 1$. The learning algorithm then minimises a **weighted empirical risk**:

$$R_{\text{weighted}}(\theta) = \frac{1}{n} \sum_{i=1}^n w_i \ell(f_{\theta}(x_i), y_i),$$

so that under-represented combinations contribute more to the optimisation objective, pushing the model towards parity.

Step-by-step toy example

Using a six-record dataset with balanced sex ($P(A = \text{male}) = P(A = \text{female}) = 0.5$) and balanced labels ($P(Y = 1) = P(Y = 0) = 0.5$), but with males having a positive rate of $2/3$ and females $1/3$, the joint frequencies are:

$$P(A = \text{male}, Y = 1) = \frac{2}{6}, \quad P(A = \text{male}, Y = 0) = \frac{1}{6},$$

$$P(A = \text{female}, Y = 1) = \frac{1}{6}, \quad P(A = \text{female}, Y = 0) = \frac{2}{6}.$$

Applying the formula gives weights $w(\text{male}, 1) \approx 0.75$, $w(\text{male}, 0) \approx 1.5$, $w(\text{female}, 1) \approx 1.5$, $w(\text{female}, 0) \approx 0.75$. The rare combinations (female & positive, male & negative) receive higher weights, increasing their influence during training.

Limitations

Reweighting can increase estimation variance when weights are large, is sensitive to small cell counts, and addresses only observational imbalance rather than causal bias. What changes is *only the sample weighting*: feature values, labels, and group membership are all preserved.

6.3.4 Comparing DIR and Reweighting

Property	DIR	Reweighting
Modifies features X	✓	×
Modifies labels Y	(✓)	×
Modifies empirical risk	indirectly	✓
Targets independence	✓	✓
Model-agnostic	✓	✓
Risk of high variance	low	high (if large weights)

The key conceptual difference is that **DIR reshapes the feature distribution** — it makes $P(X' | A = a) \approx P(X' | A = b)$ — while **Reweighting reshapes the optimisation objective** — it makes $P_w(A, Y) \approx P(A)P(Y)$.

Both methods approximate **demographic parity** (independence): $P(\hat{Y} = 1 | A = a) = P(\hat{Y} = 1 | A = b)$. In a toy numerical comparison, a baseline classifier using an income threshold of 50 achieves a demographic parity gap $\Delta_{\text{DP}} = 1.0$. After full DIR repair the gap drops to $\Delta_{\text{DP}} = 0$ (transformed incomes move the boundary to equalise group positive rates). Reweighting in the same example achieves $\Delta_{\text{DP}} = 0.5$ — a meaningful reduction, but less complete, because it modifies the optimisation objective rather than the feature geometry itself.

Importantly, there is **no universally superior mitigation method**. Performance depends on data geometry, the chosen fairness notion, the model class, and the acceptable accuracy trade-off.

6.3.5 In-processing: Prejudice Remover

Main idea

The **Prejudice Remover** regulariser, introduced by Kamishima et al. (2012), integrates

fairness directly into the learning objective by augmenting the standard prediction loss with a differentiable fairness penalty:

$$\min_{\theta} \mathcal{L}(\theta) + \lambda \Phi_{\text{fair}}(\theta).$$

The key design choices are: selecting which fairness criterion to penalise (e.g. demographic parity, equalized odds), defining a differentiable penalty function $\Phi_{\text{fair}}(\theta)$, and tuning λ to control the fairness–accuracy trade-off.

Demographic parity penalty

A natural penalty for demographic parity computes the squared difference of the mean predicted probability across protected groups:

$$\Phi_{\text{DP}}(\theta) = (\bar{p}_{A=0}(\theta) - \bar{p}_{A=1}(\theta))^2, \quad \bar{p}_{A=a}(\theta) = \frac{1}{n_a} \sum_{i: A_i=a} \hat{p}_i(\theta),$$

where $\hat{p}_i(\theta) = \sigma(x_i^\top w)$ is the model’s predicted probability for instance i . The gradient of Φ_{DP} with respect to $\hat{p}_i$ is:

$$\frac{\partial \Phi_{\text{DP}}}{\partial \hat{p}_i} = \begin{cases} 2(\bar{p}_{A=0} - \bar{p}_{A=1})/n_0 & \text{if } A_i = 0, \\ -2(\bar{p}_{A=0} - \bar{p}_{A=1})/n_1 & \text{if } A_i = 1. \end{cases}$$

If the model currently assigns systematically higher predicted probabilities to group $A = 0$, the gradient pushes those probabilities down and the group- $A = 1$ probabilities up, reducing the group mean disparity. Via the chain rule, this gradient propagates back to the model parameters w .

Step-by-step toy example

Using the same four-record dataset as before, at initialisation ($w = 0$) all predicted probabilities equal $\sigma(0) = 0.5$, so $\Phi = 0$ and the penalty contributes nothing. After a few gradient steps, suppose $\hat{p} = [0.85, 0.73, 0.25, 0.15]$. Then $\bar{p}_{A=0} = 0.79$, $\bar{p}_{A=1} = 0.20$, and $\Phi \approx 0.348$. The penalty gradient pushes group- $A = 0$ probabilities downward and group- $A = 1$ probabilities upward. After full optimisation (qualitatively), $\hat{p} \approx [0.70, 0.60, 0.55, 0.45]$, with both group means near 0.65 and Φ much closer to zero.

Typical outcomes and applicability

The Prejudice Remover yields a reduced gap in positive prediction rates between groups (targeting demographic parity), a potential accuracy loss controlled by λ , and — when λ is large — a model that may sacrifice predictive performance substantially to satisfy parity. It is appropriate when the practitioner can modify the training procedure, has access to labels, and desires an explicit, tunable fairness–accuracy trade-off.

6.3.6 In-processing: Adversarial Debiasing

Adversarial debiasing, proposed by Zhang et al. (2018), extends the standard neural network training loop with a second **adversarial network** whose task is to predict the protected attribute A from the predictions $\hat{Y}$ of the primary classifier. The total training objective simultaneously minimises the classifier’s class-prediction loss and *maximises* the adversary’s prediction loss — making it as hard as possible for the adversary to infer A

from $\hat{Y}$. This adversarial formulation provides a flexible, gradient-based mechanism for enforcing independence between predictions and the protected attribute during training.

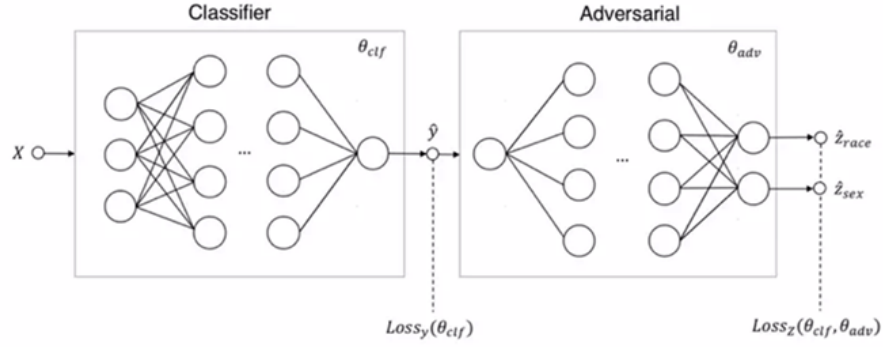


Figure 6: Diagram of adversarial debiasing architecture: input x feeds into a classifier network (parameters θ_{clf}) producing $\hat{y}$, which feeds into an adversarial network (parameters θ_{adv}) producing predictions of protected attributes (race, sex); two separate loss functions $\text{Loss}_y(\theta_{\text{clf}})$ and $\text{Loss}_z(\theta_{\text{clf}}, \theta_{\text{adv}})$ are shown